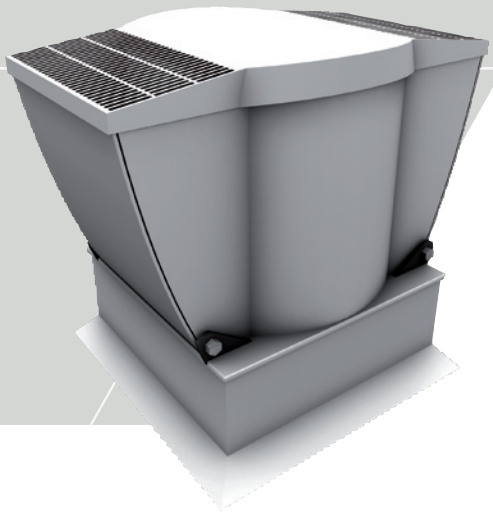
FläktGroup

Roof*JETT*

OPERATION MANUAL



Product range



Type code

Roof*JETT*

R J	V	L	.	3 1 4 0	.	4	B	2	0
Roof fans	Air-flow function	Temperature range		Pipe connections		Fan impeller	Electric motor type	Electric equipment	Electric motor speed	Colour

RJ	Roof*JETT*

Fan casing

V	Tiltable fan casing version
N	Rigid fan casing version
S	Sound damping fan casing version

Electric motor

		Temperature of air to be extracted in °C
L	Electric motor inside airflow	-20 to 70°C
M	Electric motor inside airflow	-30 to 105°C
Y	Explosion-proof version “d” (Roof fans II 2G Ex h IIB+H_2 T4 Gb)[3]	-20 to 40°C

	Ductwork [4]	**Fan impeller**
2022	Ø 200 mm	Ø 225 mm
2025	Ø 200 mm	Ø 250 mm
2528	Ø 250 mm	Ø 280 mm
2531	Ø 250 mm	Ø 315 mm
3135	Ø 315 mm	Ø 355 mm
3140	Ø 315 mm	Ø 400 mm
4045	Ø 400 mm	Ø 450 mm
4050	Ø 400 mm	Ø 500 mm
5056	Ø 500 mm	Ø 560 mm
5063	Ø 500 mm	Ø 630 mm
6371	Ø 630 mm	Ø 710 mm
6380	Ø 630 mm	Ø 800 mm

Electric equipment

0	EC Motor
2	AC motor – 2-pole – 3000 min^{-1} – 50 Hz
4	AC motor – 4-pole – 1500 min^{-1} – 50 Hz
6	AC motor – 6-pole – 1000 min^{-1} – 50 Hz

A	Fan terminal block
B	Fan switch

1	AC motor– 3 ~ 400 V– 1-speed
2	AC motor– 3 ~ 400 V– 2-speed [5]
4	AC motor– 1 ~ 230 V– 1-speed [5]
D	AC motor – 3~ 400 V – 2-speed – (YY/Y)
Z	EC motor – 3~ 400 V – continuous speed control

Colour

0	RAL 9002 - grey and white (standard)
9	RAL on request (must be specified in the order)

3 - Electric motor - Flameproof enclosure “d”CSN EN 14986 ed.2, CSN EN ISO 80079-36, CSN EN 13463-3, CSN EN IEC 60079-0 ed.5 and CSN EN 60079-1 ed.3

4 - Air duct connection CSN EN 12220

5 - Speed control possible by voltage adjustment

Product range 2

Type code 3

1 Safety and information for the user 6

1.1 Scope of the operation manual 6
1.2 Symbol 6
1.3 Safety at work 7
1.4 Use for intended purpose 8
1.5 Improper use 8
1.6 Safety regulations and standards 9
1.7 Changes and modifications 9
1.8 Spare parts 9
1.9 Staff qualifications 9
1.10 Warranty 9

2 Technical data 10

2.1 Fan components 10
2.2 Material specifications 14
2.3 Technical description 14
2.3.1 Fan properties 14
2.3.2 Fan design series 14
2.3.3 Operating conditions and technical data of electric fan motors 15
2.4 Fan dimensions 15
2.5 Minimum RJVY fan impeller blade clearance 16
2.6 Accessories 16
2.7 Technical data 17

3 Transport and storage 21

3.1 Transport safety 21
3.2 Content of delivery 21
3.3 Packaging 22
3.4 Transport 22
3.5 Storage 22
3.6 Disposal 22

4 Installation 23

4.1 Installation site 23
4.1 Installation 23
4.2.1 Roof plinth 23
4.2.2 Passage through the roof 25
4.2.3 Roof sealing 26
4.2.4 Roof fan installation 27
4.2.5 Fan earthing 29

5 Electrical connections 30

5.1 Terminal block / fan switch wiring diagrams30
5.2 Controllers and fan connection 30
5.3 Cable connections 30
5.4 Electric motor protection using thermal contacts / thermistors 31
5.5 Power supply voltage fluctuation 31
5.6 Fan speed control 31
5.6.1 Fan speed control by power supply voltage adjustment 31
5.6.2 Fan speed control by frequency inverter 31

5.7 Electrical connections 32
5.7.1 Flange mounted AC motors 32
5.7.2 AC motors with an external rotor 32
5.8 Electrical connections for a group of fans 35

6 Commissioning 38
6.1 Safety inspection 38
6.2 First commissioning 39
6.2.1 Trial run 39
6.2.2 First commissioning procedure 40

7 Operation 41
7.1 Switching the fan off 41
7.1.1 Starting the fan after maintenance or repairs 41

8 Maintenance and troubleshooting 42
8.1 Regular inspection of RJ#L and RJ#M fans 42
8.2 Regular inspection of RJVY fans 42
8.3 Operation faults 42

9 Dismantling and disposal 44
9.1 Dismantling 44
9.2 Recycling 44

Inspection 45

Declaration of conformity 46

Copyright
Distributing and copying of this document, its sale and the disclosure of its content is strictly prohibited without express prior consent. Any violation of this clause will result in liability for damages. All rights in relation to patent, utility model or industrial design registration are reserved.

1 Safety and user information

This is the original operation manual verified by the manufacturer.

RoofJETT roof fans are developed and manufactured in accordance with state-of-the-art technological standards and established technical safety standards and regulations. They comply with EN standards and the Machinery Directive.

RoofJETT roof fans ensure maximum operational safety and meet the highest quality standards. This product line combines future-oriented technology with operational and maintenance considerations.

Nevertheless, all fans pose unavoidable residual risk to the user or third parties, or cause damage to the unit or other property. For this reason, all safety rules must be observed. Failure to follow safety instructions could result in death, serious injury, environmental damage and/or considerable property damage.

Compliance with the safety instructions in this operation manual will help prevent risk and ensure economical operation and appropriate use.

1.1 Operation manual application scope

This operation manual contains information on the following areas:

– Transport and storage
– Fitting
– Installation
– Electrical connections
– Commissioning
– Operation
– Maintenance and troubleshooting

1.2 Symbols

The following symbols are used in the manual to highlight parts of the text:

– This symbol is used to indicate standard lists.
• This symbol indicates steps to take.
✓ This symbol indicates the results of steps taken.



Attention!
This symbol indicates supplementary information on the use of RoofJETT roof fans and their accessories.



Recycling
This symbol indicates information on handling packing material and used machine parts (classified according to the type of material used).

The following symbols are used throughout the manual as warnings:



Risk of electrocution!
This symbol indicates activities posing a risk of electrocution.



Injury to persons!
This symbol indicates special information, commands and restrictions to avoid injury to persons!



Risk of injury from suspended loads!
This symbol indicates a risk of injury and damage due to suspended loads.



Sharp edges can cause injuries!
This symbol indicates instructions, commands and restrictions which, if not observed, may result in cuts or damage caused by sharp edges.



Risk of injury from rotating parts!
This symbol indicates warnings, commands and restrictions which must be observed to avoid injury or damage caused by rotating parts!



Damage to the environment!
This symbol warns of damage to the environment and refers to the applicable environmental protection regulations.



Damage to the fan!
This symbol refers to special data, commands and restrictions to avoid damage to the unit.

1.3 Safety at work

To ensure your own safety, comply with the following safety instructions:



Risk of electrocution!
Disconnect the fan from the power supply and ensure that it cannot be switched back on. Earth, short-circuit and disconnect any adjacent components which are connected to the power supply. Failure to observe this procedure may lead to serious injury or death.

- When working on the fan, it must be disconnected from the power supply.
- Use protective equipment when transporting and assembling the fan.

1.4 Use for intended purpose

The unit must be used pursuant to Commission Regulation (EU) No. 1253/2014.

RoofJETT roof fans are designed to exhaust dust-free air and other non-corrosive gases. The maximum temperature of the medium to be extracted is 45°C for RJ#L series roof fans with electric motors with an external rotor, and 40°C for RJVY fans. For RJ#L fans the gas to be extracted must not contain more dust than air in industrial areas (< 5 mg/m^3).

RJ#M series roof fans with flange mount squirrel cage electric motors installed away from the air flow may operate at higher temperatures (up to 105°C). When extracting heavily contaminated air, regular maintenance is required to prevent an adverse effect on the balance of the impeller caused by dust particles adhering to or building up on the impeller surface, and to ensure that the shut-off dampers are always operational (they must open freely with the flow of air and close by gravity when the fan is switched off).

For the exact technical data on temperature, dust content and electric motor types see Table 2-2 on page 15.

A fan switch is used to deactivate the fan and prevent unintentional activation during maintenance. This switch does not act as the main or emergency switch.

If automatic control equipment with electronic components is used (e.g. frequency inverters), the manufacturer's recommendations for preventing electromagnetic interference (EMC) must be observed (earthing, cable lengths, cable shielding etc.).



Risk of injury from rotating parts!
If a roof fan is supplied without protective features, the user must make arrangements for a protective grille according to CSN EN ISO13857 (e.g. RJZ#31 - Air intake grille) to be mounted in front of the impeller.



Attention!
Any other use or use beyond this scope will be considered use not in compliance with the guidelines. The manufacturer / supplier will not be liable for any resulting damage and the risk will be borne by the user.

The user is responsible for use in compliance with the guidelines.

Intended purpose use also includes compliance with instructions contained in the operation manual and conditions for inspection and maintenance defined by FläktGroup.

1.5 Improper use

Improper use of the fan when extracting:

- air the temperature of which is too high or too low
- corrosive air
- air with a high dust content
- in corrosive environments



Injury to persons!
Improper use could result in injury to persons and damage to property, leading to bearing damage, damage due to corrosion, imbalance, vibration, deformation and damage due to abrasion. Improper use can cause the impeller or the shaft to shear, permanent fractures, fatigue fractures and fire caused by sparks.

1.6 Safety regulations and standards

When carrying out the installation, electrical connection, commissioning, maintenance and servicing of fans, all safety regulations and standards must be observed, in particular CSN 12 2002 and CSN EN 60204-1 ed.3.

- CSN 331310 ed. 2Electrical engineering regulations. Safety regulations for electrical equipment intended to be used by persons without any electrical engineering qualifications.
- CSN 332000-1 ed. 2Electric engineering regulations. Basic provisions for electrical equipment.

Fans designed for explosive environments (CSN EN ISO 80079-36) are protected against the intrusion of foreign objects by protective grilles with a protection rating of IP20 (CSN EN 60529).

1.7 Changes and modifications

Neither Roof*JETT* fans nor their components may be modified in any manner.
Any modification to the fan or its components will invalidate the warranty and declaration of conformity of the Roof*JETT* fan.

1.8 Spare parts

Only original spare parts are permitted. The manufacturer is not liable for any damage or injury if third-party spare parts are used.

1.9 Operator qualifications



Attention!
Each person assigned to work on Roof*JETT* fans must read and fully understand this operation manual.

Installation/commissioning/maintenance/repair: Only by qualified and trained HVAC professionals.

Electrical installation: Only by trained electrical engineers qualified pursuant to Clause 6 of Regulation No. 50/78 Coll. of the CUBP and CBU.

1.10 Warranty

FläktGroup shall not be held liable for any claims resulting from warranties and guarantees in the event of injury to persons and damage to property if they are related to one or more of the following causes:

- The fan was not handled in compliance with the guidelines (see Chapter 1.4).
- The installation, commissioning and operation of the fan were not carried out properly.
- The fan was operated with improperly working or improperly located safety features.
- Instructions from the operation manual were not observed.
- The fan or its parts were modified or tampered with.
- Unprofessional or untimely maintenance.

FläktGroup shall not be held liable for any subsequent damage.

2 Technical data

2.1 Fan components

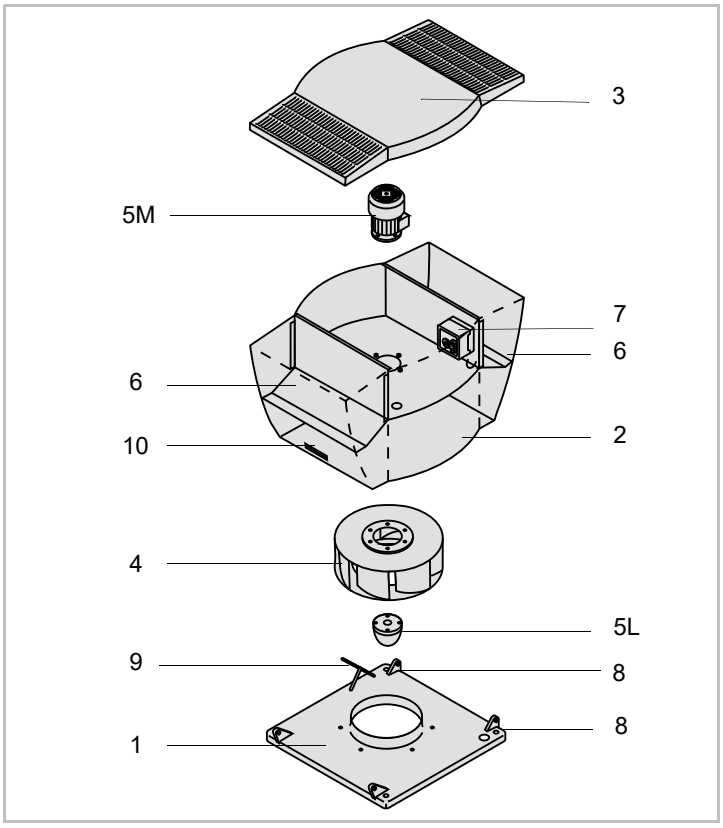


Fig. 2-1: Fan components

Pos.: 1 A base plate sloped to allow water drainage, with closed corners and a cable bushing

Pos.: 2 Fan casing optimised for air flow

Pos.: 3 Cover hood

Pos.: 4 Radial fan with backward-curved blades

Pos.: 5L Electric motor with an external rotor, acting as a fan hub

or

Pos.: 5M Flange mounted squirrel cage electric motor with armature, installed away from the air flow

Pos.: 6 Automatic shut-off dampers

Pos.: 7 Fan switch or terminal block

Pos.: 8 Hinges for tiltable fan casing

Pos.: 9 Tiltable fan casing support bracket

Pos.: 10 Tiltable fan casing handle

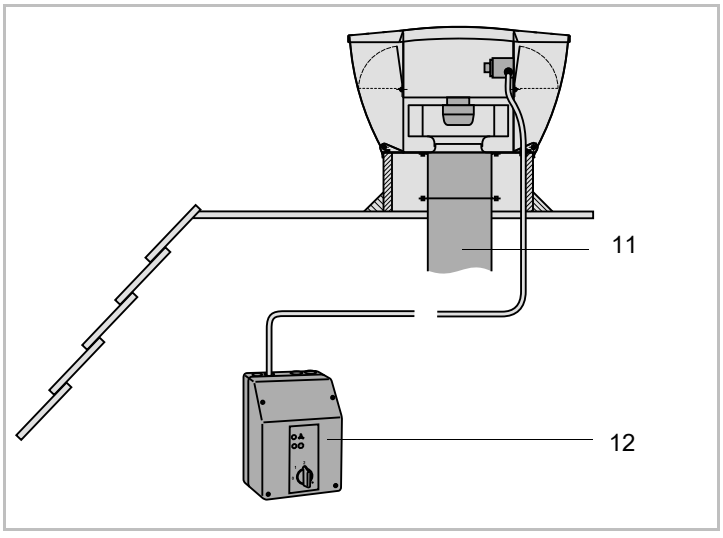


Fig. 2-2: Fan installation

Pos.: 11 Circular air duct

Pos.: 12 Fan controller

Some fan components from Fig. 2-1 and Fig. 2-2 are described in more detail below:

Base plate (pos. 1) **with hinges** (pos. 8) **and a support bracket** (pos. 9)
A sealed base plate made of galvanised steel (the RJVY type is made of aluminium), protected with a thermally cured powder coating, corrosion resistant, designed to be mounted on plinths. The base plate includes hinges to allow for the entire fan casing to be tilted to one side for inspection and maintenance. The high air intake nozzle helps to enhance the air capacity and reduces the noise level. The sloping base plate ensures that water is efficiently drained from the fan plate. The air intake nozzle is also fitted

with threaded nut holes along its circumference that can be used to connect air ducts (11) or Roof*JETT* accessories.

Fan casing (pos. 2) **with integrated shut-off dampers** (pos. 6) **and a handle** (pos. 10)

The casing is made of steel with double corrosion-resistant protection as standard – a galvanized layer and a corrosion-resistant RAL 9002 paint layer. It has two opposing air outlet channels that are directed upwards. These relatively large outlets (low air exit velocity and thus low noise) take exhaust air high over the roof in a non-rotating vertical jet.
Two lightweight automatic shut-off dampers made of aluminium are integrated into the standard casing version and remain operational in all weathers. These dampers are opened upwards by the flow of air being extracted by the fan.
When the fan stops running, the outlet ducts are closed by gravity, preventing unintentional ventilation.
A practical plastic handle is used to facilitate maintenance and service by tilting the fan casing.

Cover hood (pos. 3)

The cover hood is easy to remove for carrying out electrical connection, providing trouble-free access to the electrical enclosure. Its slightly oval shape provides efficient water drainage.

Radial fan (pos. 4)

A high-performance radial fan with an impeller with backward-curved blades, with a corrosion-resistance finish. Compared with radial fans with forward-curved blades, this design significantly reduces the chance of contamination (by dust for example). The radial fan is inside a casing and is operated by an electric motor, also inside the casing. The impeller and electric motor are dynamically balanced to level 6.3 according to CSN ISO 21940-11, (BV3 according to CSN ISO 14694).

Electric motors (pos. 5L, 5M)

The electric motor is powered by a control unit (12). For maintenance it can be switched off using the fan switch (7) in the upper part of the casing. On fans with a terminal block (7), before disconnecting the conductors in the terminal block it is necessary to switch off the unit and secure it so that it cannot be unintentionally switched back on. Speed is controlled via an external control system.

RJ#L series: In RJ#L series roof fans with AC motors or EC motors with an external rotor, the maximum temperature of the medium to be extracted is up to 70°C. In EC motors (5L) speed can be continuously controlled To do this a 0-10 V control signal is necessary. Additionally, in this series the dust content of extract air may not significantly exceed the equivalent dust concentration in outside air in industrial areas (< 5 mg/m3). AC motors with an external rotor are thermally protected by a thermal contact (TC). Fans with EC motors in the RJ#L series are available in sizes 3135, 3140, 5056 and 5063.

RJ#M series: RJ#M series roof fans with flange mount squirrel cage electric motors installed away from the air flow, are permitted to operate at higher temperatures (up to 105°C) and with air containing dust. These electric motors can be operated at temperatures between - 30°C and 60°C (at 40°C with a power factor of 1.1, using thermal class F, or higher than 40°C with reduced capacity). Electric motors can be controlled via an inverter under the following voltage conditions: ts > 0.1 ms at $U \leq 500$ V. Electric motors are thermally protected by a thermal contact (TC).

RJVY series: RJVY roof fans with explosion-proof flange mount squirrel cage electric motors installed away from the air flow can permanently extract media of temperature of up to 40°C. Electric motors are thermally protected by a thermistor.

Circular air duct (pos. 11, fig. 2-2)

Risk of injury from rotating parts!
Fit a safety grille on the air intake side (available as an accessory) in accordance with CSN EN ISO 13 857 if there is a risk of contact with the rotor.

Roof*JETT* roof fans can draw air in directly or can be connected to a circular air duct.

There are several options:

When insulated plinth RJZ#10 or insulated plastic plinth RJZ#12 are used:

The circular air duct is located in the centre of the roof plinth.

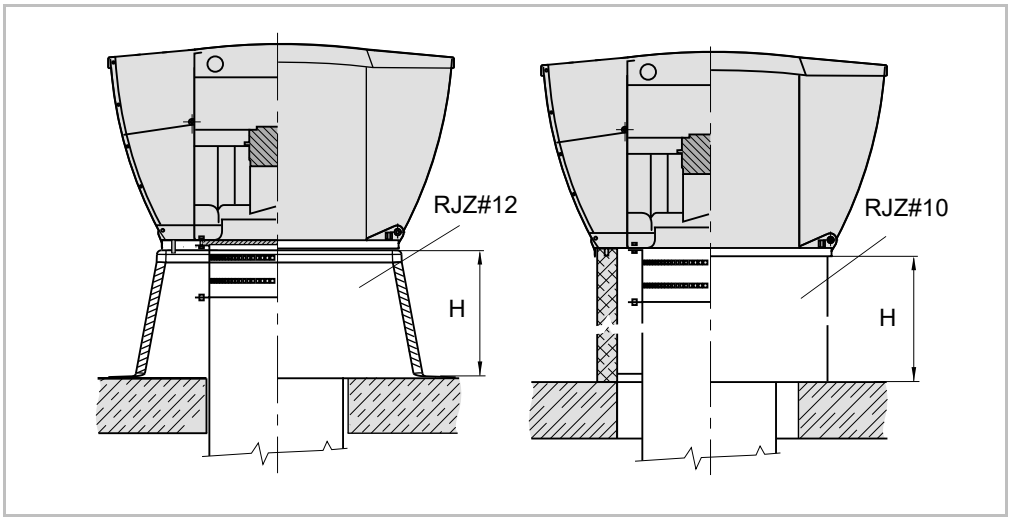
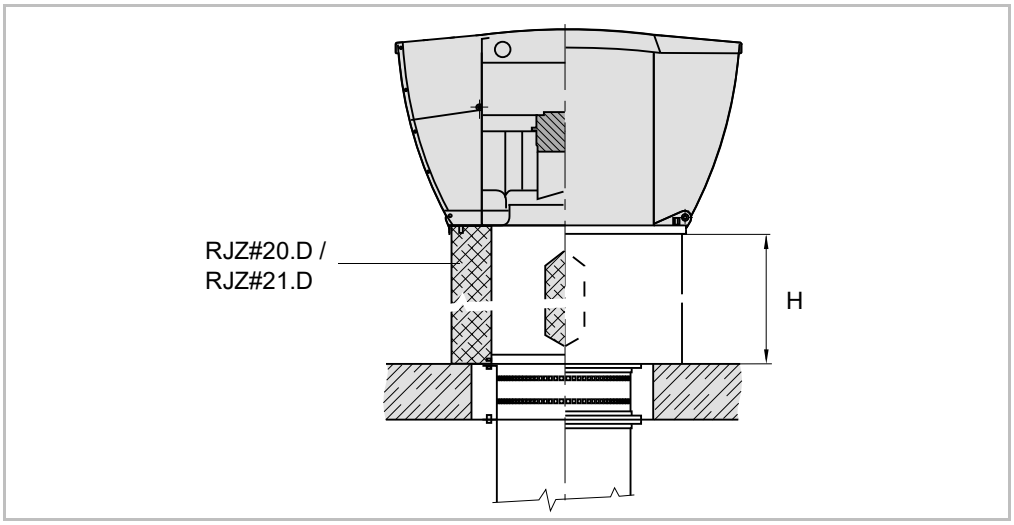


Fig. 2-3: A circular air duct in insulated plinth RJZ#10 (plinth height H = 300 mm) and in insulated plastic plinth RJZ#12 (plinth height H = 300 mm)

When sound damping plinth RJZ#20 (high installation) or RJZ#21 (low installation) is used:

A reducer plate can be used on the bottom of the roof plinth – version D – which can be connected to a circular air duct.



Pos. 1: A circular air duct in sound damping plinth RJZ#20D (H=700 or 950 mm) or RJZ#21.D (H = 370 or 500 mm) and with a reducer plate

When insulated plinth RJZ#10 or sloped plinth RJZ#11 with sound damping insert RJZ#22 are used:

The sound damping insert is fitted with a reducer plate so that an air duct can be connected to the bottom of the insert.

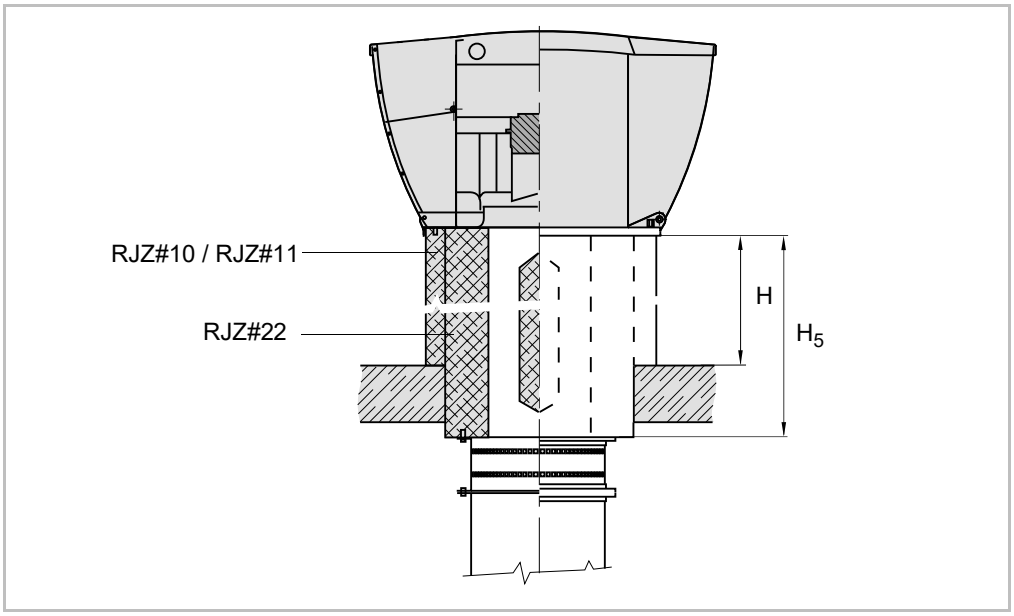


Fig. 2-4: An air duct with sound damping insert RJZ#22 (H5 = 850 mm) with a reducer plate and with insulated plinth RJZ#10 (H = 300 mm) or sloped plinth RJZ#11 (H = 750 mm or 1000 mm)

Controller (pos. 12, fig. 2-2)

Roof*JETT* roof fans can be operated using a FläktGroup controller or an external control system.

FläktGroup controller / control system options:

- Switch MC301/MC331
- Five-stage control transformers
- Phase-fired control systems

All Roof*JETT* roof fans are supplied with connections from the factory.

There are 12 installation sizes available.

2.2 Material specifications

Fan parts	Material
Base plate	Powder-coated galvanized steel or aluminium
Fan casing	Powder-coated galvanized steel or aluminium
Cover hood	Powder-coated galvanized steel or aluminium
Fan impeller	Aluminium
Electric motor	More kinds of material
Shut-off dampers	Aluminium
Bushings	Polyester
Handle	Polyester
Connecting cable	Copper
PVC hose	Polyethylene
Terminal block	Polyester
Fan switch	Polyester

Table 2-1: Fan material specifications

2.3 Technical description

Roof*JETT* roof fans are generally designed to extract dust-free non-corrosive air.

Roof*JETT* roof fans have various applications including

- Air extraction from processing and industrial facilities and warehouses
- Air extraction from sanitary facilities in office building and multiple dwelling buildings
- Night cooling and airing of façades

2.3.1 Fan properties

- Flow rates of up to 19,160 m^3/h
- Integrated shut-off dampers in standard types
- Optimum supply and exhaust air flow
- Double corrosion protection (Class C4 - high - pursuant to CSN EN ISO 12944-2)

2.3.2 Fan configurations

The following fan configurations are available:

RJV# **Roof*JETT* vertical**

A radial roof fan with vertical exhaust airflow, with integrated automatic shut-off dampers and a tiltable fan casing.

RJN# **Roof*JETT* vertical**, **“No Swing-up” design**

RJV#, without the tiltable fan casing

RJS# **Roof*JETT* vertical, “Silent”**

A standard RJV# type with sound attenuation in the fan casing.

RJVY **Roof*JETT* vertical**

Explosion-proof version II 2G Ex h IIB+H_2 T4 Gb

2.3.3 Operating conditions and technical data of electric fan motors

Technical data	RJ#L		RJ#M	RJVY
Electric motor type	AC	EC	AC	AC
Min. extraction air temperature	- 25°C [1)]	- 25°C [1)]	- 30°C	- 20°C
Max. extract air temperature	40 to 70°C [1)]	60 to 70°C [1)]	up to 105°C	40°C
Max. dust load	5 mg/m	5 mg/m	10 mg/m	10 mg/m
Min. installation temperature	- 25°C	- 25°C	- 30°C	- 20°C
Max. installation temperature	45°C	40°C	40°C	40°C
Electric motors				
Electric motor version	with an external rotor	with an external rotor	Flange mounted squirrel cage	Flange mounted squirrel cage
Electric motor installed inside airflow	YES	YES	NO	NO
Protection rating	IP44 / IP 54	IP 54	IP 55	IP 54
Temperature class	F	F	F	F
Max. air RH	100 %	100 %	100 %	100 %
Explosion-proof type	NO	NO	NO	Explosion-proof “d”

1) according to electric fan motor type

Table 2-2: Operating conditions and technical data of electric fan motors

2.4 Fan dimensions

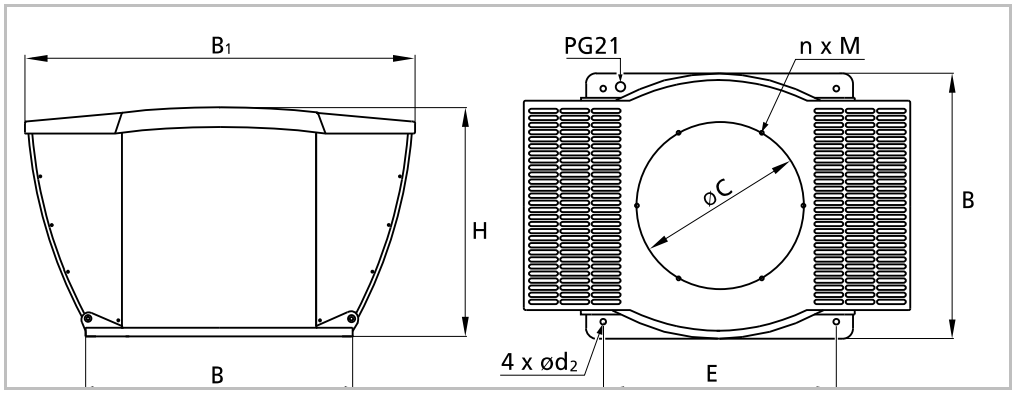
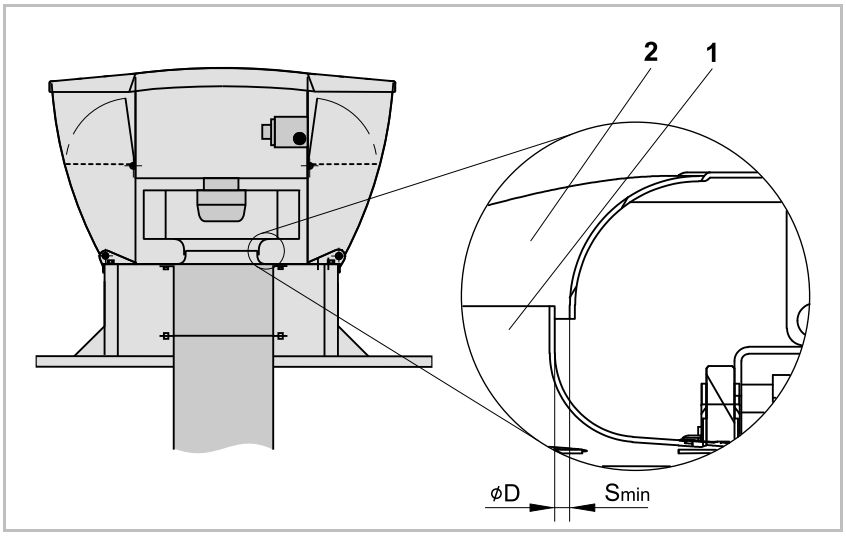


Fig. 2-5: Dimensions

Fan size	B_1 [mm]	H [mm]	□B [mm]	□E [mm]	4x ø d_2 [mm]	ø C [mm]	n x M [mm]	S min [mm]	Weight [kg]
2022	588	364	400	330	9	233	6 x M8	2	Approx. 18
2025	588	364	400	330	9	233	6 x M8	2	Approx. 21
2528	588	364	400	330	9	283	6 x M8	2	Approx. 20
2531	820	480	560	490	12	283	6 x M8	2	Approx. 33
3135	820	480	560	490	12	352	8 x M8	2.3	Approx. 33
3140	820	480	560	490	12	352	8 x M8	2.6	Approx. 36
4045	1168	632	800	730	14	438	8 x M8	2.9	Approx. 69
4050	1168	632	800	730	14	438	8 x M8	3.3	Approx. 76
5056	1168	632	800	730	14	538	8 x M8	3.7	Approx. 73
5063	1630	865	1120	1050	14	538	8 x M8	4.1	Approx. 154
6371	1630	865	1120	1050	14	670	12 x M8	4.6	Approx. 178
6380	1630	865	1120	1050	14	670	12 x M8	5.2	Approx. 198

Table 2-3: Dimensions and weights for all installation sizes

2.5 Minimum RJVY fan impeller blade clearance



Pos. 1: Base plate
Pos. 2: Fan impeller

For min. fan impeller blade clearance S_{min} see Table 2-3 on page 15.

Fig. 2-6: Minimum fan impeller blade clearance

2.6 Accessories

The following accessories can be supplied with a Roof*JETT* roof fan

Roof plinth, sound attenuator, circular air duct connection

Roof plinth	**Sound attenuator**
Insulated plinth RJZ#10	Sound damping plinth RJZ#20.D, for connection to air duct
Sloped plinth RJZ#11	Sound damping plinth RJZ#20.F, free - without fittings for connecting air duct
Insulated plastic plinth RJZ#12	Sound damping plinth RJZ#20.D low, for connection to air duct
	Sound damping plinth RJZ#20.F, free - without fittings for connecting air duct
	Sound damping insert RJZ#22 (to be mounted in RJZ#10 or RJZ#11)[1)]
Circular air duct connection	
Air intake grille RJZ#31	Flanged air intake nozzle RJZ#35
Flexible adaptor RJZ#32	Shut-off damper with actuator RJZ#36
Rigid adaptor RJZ#33 (length 500 mm)	Top insert RJZ#37
Flange RJZ#34	Bottom insert RJZ#38

[1)] to be retrofitted on plinths without sound attenuators

Table 2-4: Accessories

2.7 Technical data

Roof fan type	Speed stage	El. motor speed	Max. air flow rate	Max. extraction air temperature	Power input	Rated current	SFP	Voltage / frequency	El. motor protection TC- thermal contact TCN-thermal contact not wired	Protection rating	Exhaust sound power level L_{wa3}	Intake sound power level L_{wa8}	Exhaust sound pressure level L_{pa3}	Intake sound pressure level L_{pa8}
	-	min^{-1}	m^3/h	°C	kW	A	$kW/m^3/s$	V/Hz	-	-	dB(A)	dB(A)	dB(A)	dB(A)
RJ#L (max. 40°C #. 70°C), el. motor with an external rotor inside extraction airflow														
RJ#L.2022.4#4#	1	1430	510	+50	0.04	0.20	0.28	1~ 230/50	TCN	IP 44	64	61	42	48
RJ#L.2022.2#4#	1	2500	940	+50	0.15	0.68	0.57	1~ 230/50	TCN	IP 44	80	76	58	64
RJ#L.2025.4#4#	1	1360	790	+40	0.05	0.25	0.23	1~ 230/50	TC	IP 44	66	65	44	52
RJ#L.2528.4#4#	1	1370	1230	+50	0.12	0.60	0.35	1~ 230/50	TC	IP 44	68	66	46	54
RJ#L.2528.4#2#	2	1330	1190	+70	0.07	0.13	0.21	3~ 400/50	TC	IP 44	67	65	45	53
	1	1050	950		0.05	0.08	0.19				61	59	39	47
RJ#L.2531.4#4#	1	1310	1710	+50	0.15	0.66	0.32	1~ 230/50	TC	IP 44	72	69	50	57
RJ#L.3135.4#2#	2	1340	2650	+70	0.26	0.49	0.35	3~ 400/50	TC	IP 54	77	75	55	63
	1	1060	2100		0.19	0.30	0.33				71	69	49	57
RJ#L.3135.0#Z#*	min.	1070	1930	+60	0.11	0.34	0.21	3~ 400/50	TC	IP 54	67	63	45	50
	max.	2520	4670		1.27	2.04	0.98				90	85	68	72
RJ#L.3140.4#2#	2	1340	3610	+40	0.46	0.85	0.46	3~ 400/50	TC	IP 54	79	76	57	64
	1	1060	2850		0.31	0.49	0.39				73	70	51	58
RJ#L.3140.0#Z#*	min.	890	2220	+60	0.11	0.35	0.18	3~ 400/50	TC	IP 54	66	63	44	50
	max.	2110	5400		1.37	2.27	0.91				89	84	68	72
RJ#L.4045.4#2#	2	1360	4760	+40	0.76	1.55	0.57	3~ 400/50	TC	IP 54	83	80	61	68
	1	1110	3370		0.54	0.92	0.58				76	74	54	62
RJ#L.4050.4#2#	2	1400	7000	+45	1.30	2.70	0.67	3~ 400/50	TC	IP 54	87	84	65	72
	1	1230	5410		1.00	1.60	0.67				81	79	59	66
RJ#L.5056.6#2#	2	880	6270	+40	0.65	1.40	0.37	3~ 400/50	TC	IP 54	78	75	56	63
	1	720	3970		0.44	0.80	0.40				70	67	48	55
RJ#L.5056.0#Z#*	min.	530	3510	+40	0.58	1.15	0.59	3~ 400/50	TC	IP 54	64	62	42	49
	max.	1230	8230		1.50	2.50	0.66				86	84	64	72
RJ#L.5063.0#Z#*	min.	540	5790	+60	0.20	0.51	0.12	3~ 400/50	TC	IP 54	69	67	47	56
	max.	1230	13450		2.73	4.20	0.73				90	88	68	76

** EC motors may be operated with a control voltage from a minimum of 4.5 V to a maximum of 10 V*

Roof fan type	Speed stage	El. motor speed	Max. air flow rate	Max. extraction air temperature	Power input	Rated current	SFP	Voltage / frequency	El. motor protection TC- thermal contact TCN-thermal contact not wired	Protection rating	Exhaust sound power level L_{wa3}	Intake sound power level L_{wa8}	Exhaust sound pressure level L_{pa3}	Intake sound pressure level L_{pa8}
	-	min^{-1}	m^3/h	°C	kW	A	$kW/m^3/s$	V/Hz	-	-	dB(A)	dB(A)	dB(A)	dB(A)
RJSL (max. 40°C #. 70°C), el. motor with an external rotor inside extraction airflow, sound damping fan casing														
RJSL.2022.4#4#	1	1430	470	+50	0.04	0.20	0.31	1~ 230/50	TCN	IP 44	58	55	37	43
RJSL.2022.2#4#	1	2500	870	+50	0.15	0.68	0.62	1~ 230/50	TCN	IP 44	74	70	52	58
RJSL.2025.4#4#	1	1400	690	+40	0.05	0.25	0.26	1~ 230/50	TC	IP 44	59	58	37	45
RJSL.2528.4#4#	1	1370	980	+50	0.12	0.60	0.44	1~ 230/50	TC	IP 44	61	59	40	47
RJSL.2528.4#2#	2	1330	950	+70	0.07	0.13	0.27	3~ 400/50	TC	IP 44	61	59	39	46
	1	1050	760		0.05	0.08	0.24				55	52	33	40
RJSL.2531.4#4#	1	1310	1660	+50	0.15	0.66	0.33	1~ 230/50	TC	IP 44	67	64	46	51
RJSL.3135.4#2#	2	1340	2520	+70	0.26	0.49	0.37	3~ 400/50	TC	IP 54	71	69	49	56
	1	1060	1990		0.19	0.30	0.34				65	63	43	50
RJSL.3135.0#Z#*	min.	1070	1840	+60	0.11	0.34	0.22	3~ 400/50	TC	IP 54	61	62	39	49
	max.	2520	4440		1.27	2.04	1.03				83	83	61	71
RJSL.3140.4#2#	2	1340	3330	+40	0.46	0.85	0.50	3~ 400/50	TC	IP 54	73	70	51	58
	1	1060	2640		0.31	0.49	0.42				66	64	44	51
RJSL.3140.0#Z#*	min.	890	2060	+60	0.11	0.35	0.19	3~ 400/50	TC	IP 54	62	62	40	49
	max.	2110	5030		1.37	2.27	0.98				83	83	61	71
RJSL.4045.4#2#	2	1360	4600	+40	0.76	1.55	0.59	3~ 400/50	TC	IP 54	77	75	55	63
	1	1110	3260		0.54	0.92	0.60				71	69	49	57
RJSL.4050.4#2#	2	1400	6660	+45	1.30	2.70	0.70	3~ 400/50	TC	IP 54	81	79	59	66
	1	1230	5150		1.00	1.60	0.70				76	73	54	61
RJSL.5056.6#2#	2	880	5820	+40	0.65	1.40	0.40	3~ 400/50	TC	IP 54	71	68	49	56
	1	720	3970		0.44	0.80	0.40				64	61	42	49
RJSL.5056.0#Z#*	min.	530	3310	+40	0.58	1.15	0.63	3~ 400/50	TC	IP 54	57	59	35	47
	max.	1230	7770		1.50	2.50	0.69				78	81	56	68
RJSL.5063.0#Z#*	min.	540	5425	+60	0.20	0.51	0.13	3~ 400/50	TC	IP 54	61	63	39	51
	max.	1230	12610		2.73	4.20	0.78				82	85	60	73

** EC motors may be operated with control voltage from a minimum of 4.5 V to a maximum of 10 V*

Roof fan type	Speed stage	El. motor speed	Max. air flow rate	Max. extraction air temperature	Power input	Rated current	SFP	Voltage / frequency	El. motor protection TC- thermal contact.	Protection rating	Exhaust sound power level L_{wa3}	Intake sound power level L_{wa8}	Exhaust sound pressure level L_{pa3}	Intake sound pressure level L_{pa8}
	-	min^{-1}	m^3/h	°C	kW	A	$kW/m^3/s$	V/Hz	-	-	dB(A)	dB(A)	dB(A)	dB(A)
RJ#M (max. 105°C), flange mounted el. motor away from the extraction airflow														
RJ#M.2528.4#1#	1	1370	1160	+105	0.09	0.35	0.28	3~ 400/50	TC	IP 55	68	66	46	54
RJ#M.2531.4#1#	1	1370	1740	+105	0.09	0.35	0.19	3~ 400/50	TC	IP 55	73	70	51	58
RJ#M.3135.4#1#	1	1430	2670	+105	0.25	0.72	0.34	3~ 400/50	TC	IP 55	77	75	55	63
RJ#M.3140.4#1#	1	1400	3670	+105	0.37	0.98	0.36	3~ 400/50	TC	IP 55	80	77	58	65
RJ#M.4045.4#1#	1	1420	5300	+105	0.75	1.96	0.51	3~ 400/50	TC	IP 55	85	83	63	71
RJ#M.4050.4#1#	1	1430	7330	+105	1.10	2.50	0.54	3~ 400/50	TC	IP 55	88	86	66	73
RJ#M.5056.6#1#	1	920	6590	+105	0.55	2.12	0.30	3~ 400/50	TC	IP 55	79	76	57	64
RJ#M.5063.6#1#	1	950	9200	+105	1.10	3.02	0.43	3~ 400/50	TC	IP 55	83	81	61	68
RJ#M.6371.6#1#	1	965	13480	+105	2.20	5.20	0.59	3~ 400/50	TC	IP 55	88	86	66	74
RJ#M.6380.6#1#	1	970	19160	+105	4.00	8.70	0.75	3~ 400/50	TC	IP 54	91	89	69	76
RJSM (max. 105°C), flange mounted el. motor away from the extraction airflow, sound damping fan casing														
RJSM.2528.4#1#	1	1370	1160	+105	0.09	0.35	0.28	3~ 400/50	TC	IP 55	61	59	39	47
RJSM.2531.4#1#	1	1370	1740	+105	0.09	0.35	0.19	3~ 400/50	TC	IP 55	67	64	45	52
RJSM.3135.4#1#	1	1430	2670	+105	0.25	0.72	0.34	3~ 400/50	TC	IP 55	71	69	49	57
RJSM.3140.4#1#	1	1400	3670	+105	0.37	0.98	0.36	3~ 400/50	TC	IP 55	74	71	52	59
RJSM.4045.4#1#	1	1420	5300	+105	0.75	1.96	0.51	3~ 400/50	TC	IP 55	80	77	58	65
RJSM.4050.4#1#	1	1430	7330	+105	1.10	2.50	0.54	3~ 400/50	TC	IP 55	83	80	61	68
RJSM.5056.6#1#	1	920	6590	+105	0.55	2.12	0.30	3~ 400/50	TC	IP 55	73	70	51	57
RJSM.5063.6#1#	1	950	9200	+105	1.10	3.02	0.43	3~ 400/50	TC	IP 55	77	75	56	63
RJSM.6371.6#1#	1	965	13480	+105	2.20	5.20	0.59	3~ 400/50	TC	IP 55	82	80	60	67
RJSM.6380.6#1#	1	970	19160	+105	4.00	8.7	0.75	3~ 400/50	TC	IP 54	86	84	64	71

Roof fan type	Speed stage	El. motor speed	Max. air flow rate	Max. extraction air temperature	Power input	Rated current	SFP	Voltage / frequency	El. motor protection TC- thermal contact TM-Thermistor	Protection rating	Exhaust sound power level L_{wa3}	Intake sound power level L_{wa8}	Exhaust sound pressure level L_{pa3}	Intake sound pressure level L_{pa8}
	-	min^{-1}	m^3/h	°C	kW	A	$kW/m^3/s$	V/Hz	-	-	dB(A)	dB(A)	dB(A)	dB(A)
RJVY (II 2G Ex h IIB+H_2 T4 Gb, max. 40°C), flange mounted el. motor away from the extraction airflow, explosion-proof "d"														
RJVY.2531.4#1#	1	1360	1640	+40	0.25	0.77	0.55	3~ 400/50	TM	IP 54	73	70	51	58
RJVY.4045.4#1#	1	1380	5260	+40	0.75	1.90	0.51	3~ 400/50	TM	IP 54	84	82	62	70
RJVY.5056.6#1#	1	910	6475	+40	0.55	1.60	0.31	3~ 400/50	TM	IP 54	79	76	57	63
RJVY.5063.6#1#	1	935	9000	+40	1.10	2.90	0.44	3~ 400/50	TM	IP 54	83	81	61	69

3 Transport and storage

Attention!
Transport may only be carried out by trained personnel following this operation manual and the applicable regulations.

3.1 Transport safety

Risk of injury from suspended loads!
Make sure there are no people underneath suspended loads!
Moving underneath such loads can be fatal.

Damage to the fan!
Roof*JETT* roof fans must be transported with care!
Improper transport may damage the fan. If damage occurred due to impact or falls, check the fan for proper functionality and check whether its technical data correspond to the data on the type plate.

3.2 Content of the delivery

Immediately upon delivery remove the packaging and check whether the unit has been damaged in transit (if damage is found, report this immediately to the forwarder) and check that the delivery is complete and correct. While doing so, compare the data on the fan's type plate with the data in the delivery note. This also applies to accessories.

Content of the delivery of standard versions:

– Roof*JETT* roof fan
– 4 nuts, 4 plastic caps, 4 plastic washers
– 1 sealing tape
– 1 flexible PVC hose (part of the fan)

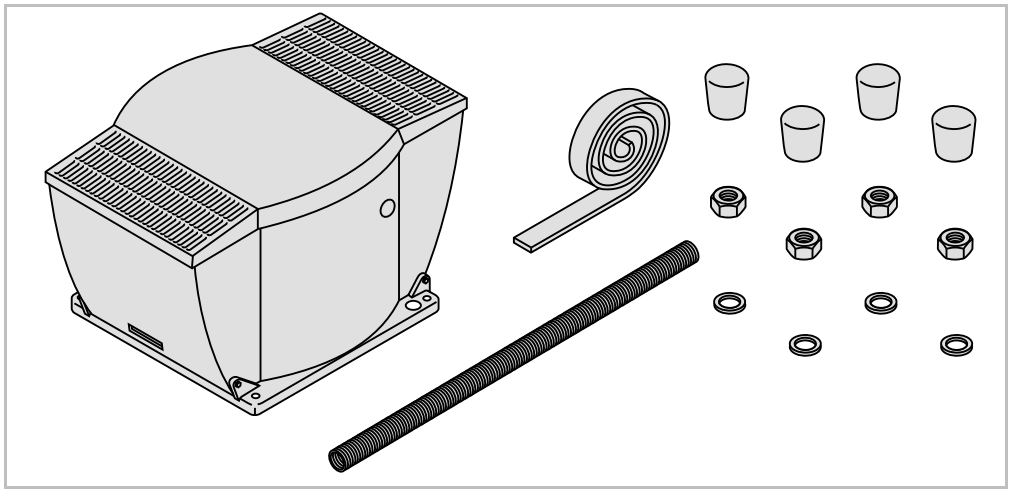


Fig. 3-1: Content of the delivery

3.3 Packaging

Fans of up to 2528 (weight including packaging of up to 25 kg) are supplied in a sturdy shipping carton. Depending on its size, the fan can be moved by one or more people using the holds provided in the packaging. Fans from size 2531 (total weight including packaging >35 kg) are supplied secured on a pallet. This can then be handled using a suitable lifting device.

3.4 Transport

It is also possible to remove the lid of the cardboard carton and take the fan out. Roof fans can only be secured for transport using their base plate or the 4 circular eyebolts, which are easily accessible after removing the cover (3). The angle of slope β must not exceed 60° (see Fig. 3-2, right).

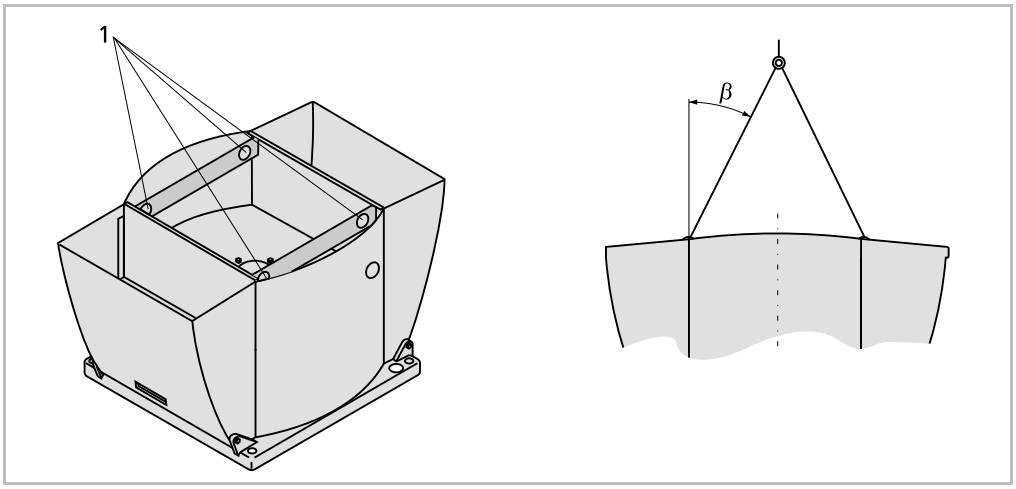


Fig. 3-2: Transport eyebolts for lifting the fan
Pos. 1: 4 transport eyebolts

Also ensure that the load is distributed evenly.



Damage to the fan!
Use a suitable means of transport for the fan type and packaging (see the data on the type plate).

3.5 Storage

The following points must be observed during storage:

– Store the fan in its original or transport packaging. This may need supplementing, depending on environmental conditions.
– The fans must be stored in a IE 12 warehouse in compliance with CSN EN IEC 60721-3-1 ed.2 protected against the elements, in a dry and dust-free environment with low relative humidity (up to 70%).
  The storage temperature must be between -30 and 40°C.,
– The fans must be protected against impact, vibration etc.

3.6 Disposal



Recycling
Operating and auxiliary substances, packaging material and parts to be replaced must be disposed of in a safe and environmentally friendly manner. In doing so, local recycling facilities must be used and local recycling regulations followed.

4 Installation

Attention!
Installation may only be carried out by trained personnel following this operation manual and the applicable regulations.
Before installation it is necessary to inspect the fan for any damage or deformation.

4.1 Installation site

The installation site must be suitable for the type, nature and ambient temperature of the fan (see Chapter 1.4 and Chapter 1.5). The following points must be observed:

- The support systems must be suitable to bear the weight of the fan, including accessories.
- The installation surface should be horizontal if possible. When the insulated roof plinth is used, the inclination of the fan must not exceed 25° if both of the opposite exhaust outlets are arranged across the slope of the roof (see Fig. 4-1).

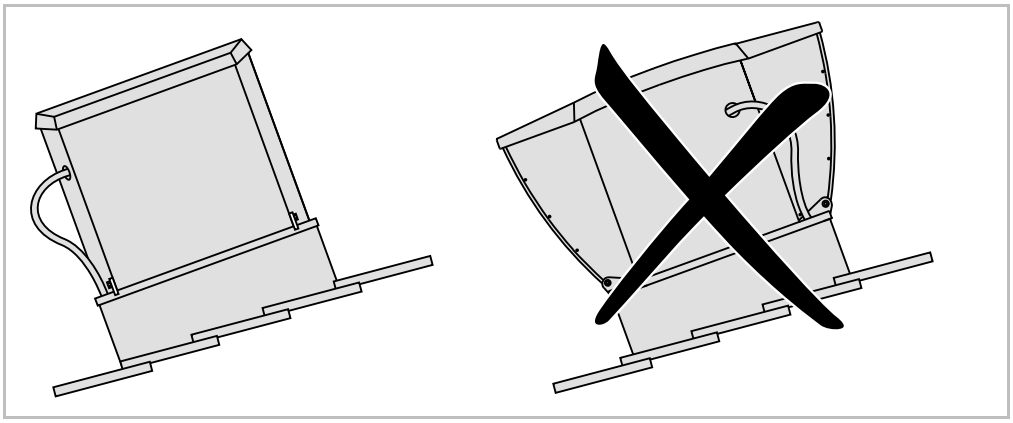


Fig. 4-1: Fan installation

- On steeply sloping roofs it is possible to level the slope by using a sloped roof plinth.
- Roof fans must be always accessible for maintenance and must be provided with sufficient clearance around them.

4.2 Installation

4.2.1 Roof plinth

Roof*JETT* roof fans are designed to be mounted on plinths.

The following plinth options are available:

- Insulated plinth RJZ#10 (for slopes up to 25°)
- Insulated plastic plinth RJZ#12 (for slopes up to 10°)
- Sloped plinth RJZ#11 with adjustable mounting brackets (up to 35 - 45° depending on the size)
- Sound damping plinth RJZ#20 (sound damping capacity up to 12 dB (A))
- Sound damping plinth RJZ#20, low (sound damping capacity up to 8 dB (A))

Plinths are available for all 12 fan types. To facilitate fan installation all roof plinths have pre-mounted bolts in the corners. They also have an integrated pipe or hole for routing connection cabling.

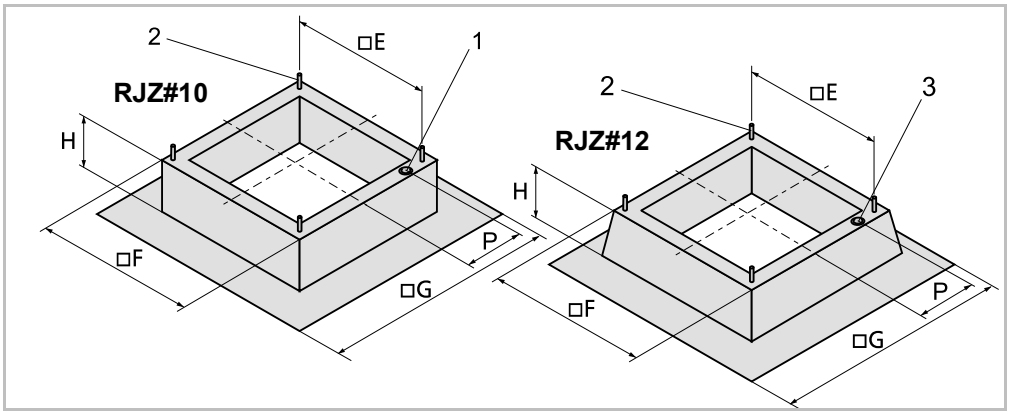


Fig. 4-2: Insulated plinth RJZ#10, plastic insulated plinth RJZ#12
Pos. 1: Free pipe for cabling integrated into the plinth wall (Ø 25mm)
Pos. 2: 4x bolts (K1)
Pos. 3: Hole for cabling (Ø 25mm)

Size	**□E** [mm]	**□F** [mm]	**□G** [mm]	**H** [mm]	**P** [mm]	**4 x K1** [mm]	**Weight RJZ#10/ RJZ#12** [kg]
2022/ 2025/2528	330	376	700	300	133	M6	7 / 6
2531 / 3135 / 3140	490	536	860	300	185	M10	12 / 8
2531 / 3135 / 3140	730	775	1100	300	305	M12	19 / 10
2531 / 3135 / 3140	1050	1095	1420	300	465	M12	27 / 13

Table 4-1: Dimensions of insulated plinth RJZ#10 / RJZ#12

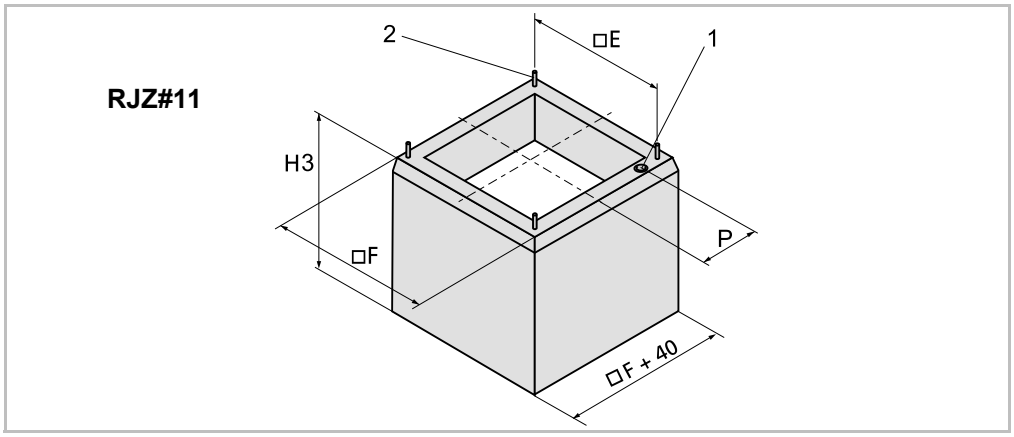


Fig. 4-3: Sloped plinth RJZ#11
Pos. 1: Free pipe for cabling integrated into the plinth wall (Ø 25mm)
Pos. 2:4x bolts (K1)

Size	**□E** [mm]	**□F** [mm]	**H3** [mm]	**P** [mm]	**4 x K1** [mm]	**Weight RJZ#11** [kg]
2531 / 3135 / 3140	330	376	750	133	M6	20
2531 / 3135 / 3140	490	536	750	185	M10	37
2531 / 3135 / 3140	730	775	1000	305	M12	66
2531 / 3135 / 3140	1050	1095	1000	465	M12	97

Table 4-2: Dimensions of insulated plinth RJZ#11 for all installation sizes

4.2.2 Passage through the roof

Attention!
The passage through the roof must be designed in consultation with an architect, designer or structural engineer.

The following dimensions must be observed:

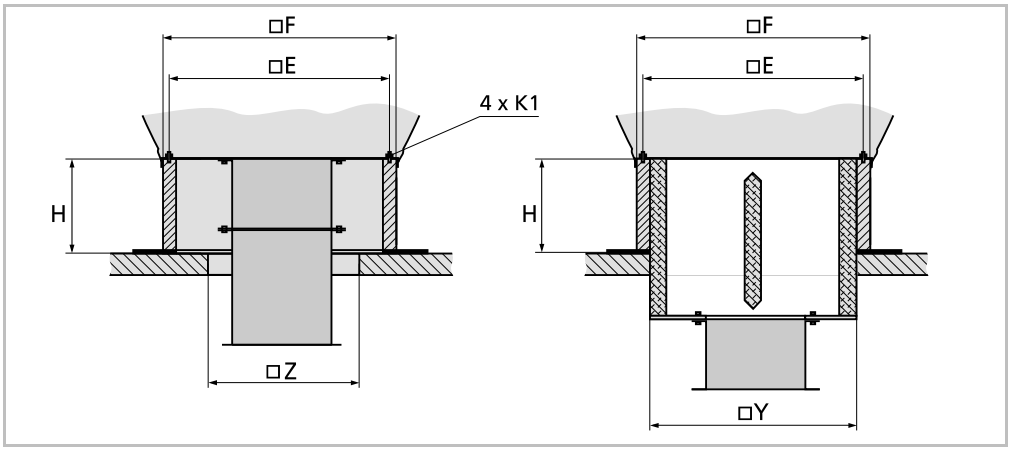


Fig. 4-4: Passage through the roof for connecting air ductwork (left) and for sound damping insert RJZ#22 (right)

Size	**□E** [mm]	**□F** [mm]	**H** [mm]	**4xK1** [mm]	**□Y**[1] [mm]	**□Z**[2] [mm]
2022 / 2025	330	376	300	M8	310	270
2528	330	376	300	M8	310	320
2531	490	536	300	M10	470	320
3135 / 3140	490	536	300	M10	470	385
4045 / 4050	730	775	300	M12	710	475
5056	730	775	300	M12	710	575
5063	1050	1095	300	M12	1030	575
6371 / 6380	1050	1095	300	M12	1030	715

[1] *If sound damping insert RJZ22 is used, the size of the opening is to be □Y*
[2] *If the roof passage is used for air ductwork, the size of the opening is to be □Z*

Table 4-3: Dimensions for passage through the roof

4.2.3 Roof seal

The gap between the roof and the plinth must be sealed by qualified specialists as follows:

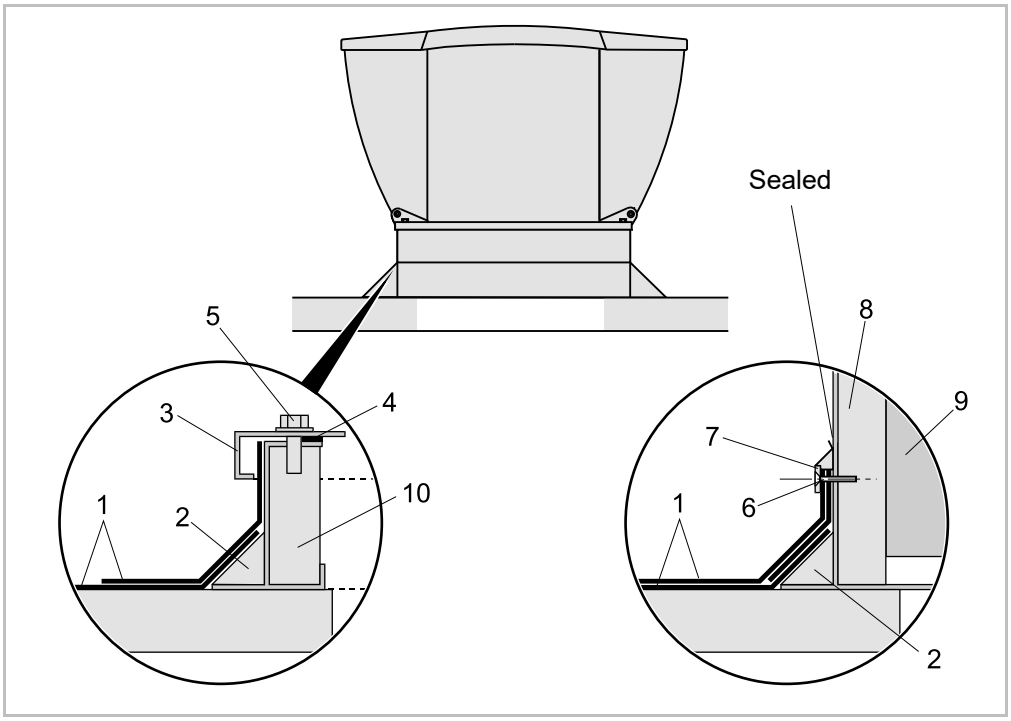


Fig. 4-5: Installation example: A RoofJETT roof fan mounted on insulated plinth RJZ#10 (left) or sound damping plinth RJZ#20 (right)

Pos. 1: Roof surface (insulation)
Pos. 2: Support wedge
Pos. 3: Fan base plate
Pos. 4: Seal (included)
Pos. 5: Nuts (included)
Pos. 6: Mounting bolt
Pos. 7: Seal profile
Pos. 8: Sound damping plinth RJZ#20
Pos. 9: Sound damping plate
Pos. 10: Insulated plinth RJZ#10

Roof seal with an insulated plinth:

With insulated plinths *RJZ#10, RJZ#12 or low sound damping plinth RJZ#21, a* seal is provided by extending the surface of the roof insulation under the fan base plate as shown in Fig. 4-5 on the left.

Roof seal with a sound damping plinth:

With sound damping plinth *RJZ#20* the surface of the roof insulation is also extended all the way to the fan base plate to prevent any problems. Alternatively, if that is not possible, the surface of the roof insulation can be secured to the roof plinth using sealing profile RJZ#13 (Fig. 4-5 right).

4.2.4 Roof fan Installation

Installation is to be carried out as follows:

Apply the seal tape supplied to the contact surface of the roof plinth (as shown in Fig. 4-6). This single-sided adhesive seal tape prevents the ingress of water and air between the fan base plate and the plinth.

Attention!
In order to seal the plinth with the fan base plate using the seal tape the plinth surface must be smooth and clean.

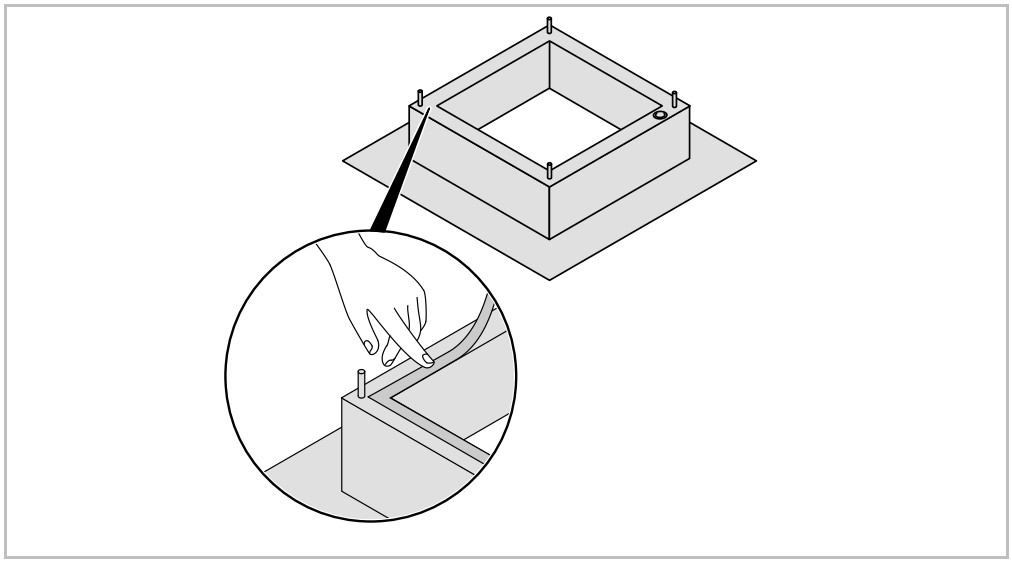


Fig. 4-6: Seal tape application

Before attaching the fan and its base plate to the plinth, we recommend that any accessories should be mounted onto the fan base plate (from the below).

Attention!
To prevent condensation on the inside of the base plate we recommend that the base plate be insulated.

Installation is to be carried out using the following steps:

1) Accessories are mounted onto the fan base plate from below. The base plate has dedicated holes for this purpose. There are 6, 8 and 12 holes depending on the type of accessory.

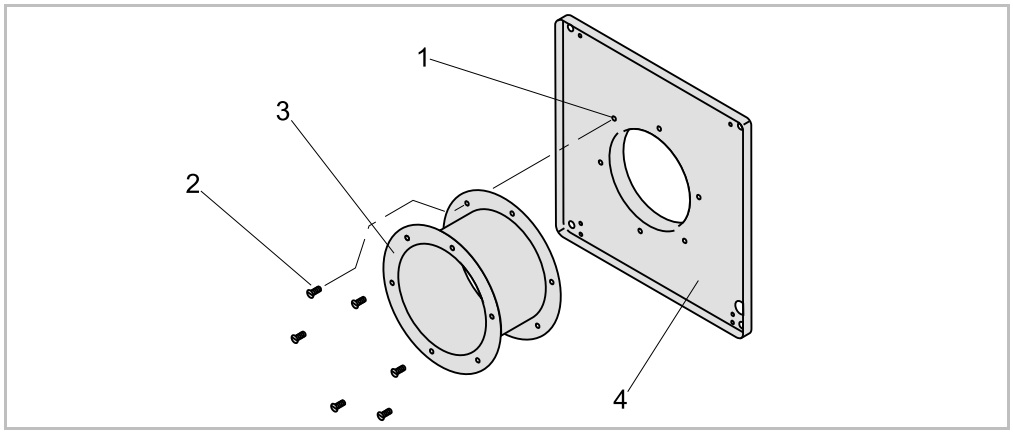


Fig. 4-7: Mounting accessories onto the fan base plate
Pos. 1: Nuts in the fan base plate
Pos. 2:Bolts
Pos. 3:Air intake adaptor with flange
Pos. 4:Fan base plate

2) Mount the roof fan with base plate and accessories on the plinth using suitable lifting equipment. Lift the roof fan using lifting eyebolts (see Fig. 3-2).

3) Mount the base plate on the bolts provided in the plinth. Screw it into place from the top using the nuts provided (the bolt size depends on the type of accessory, see tab. 4-1 to tab. 4-3) and cover them with plastic caps Use the washers provided.

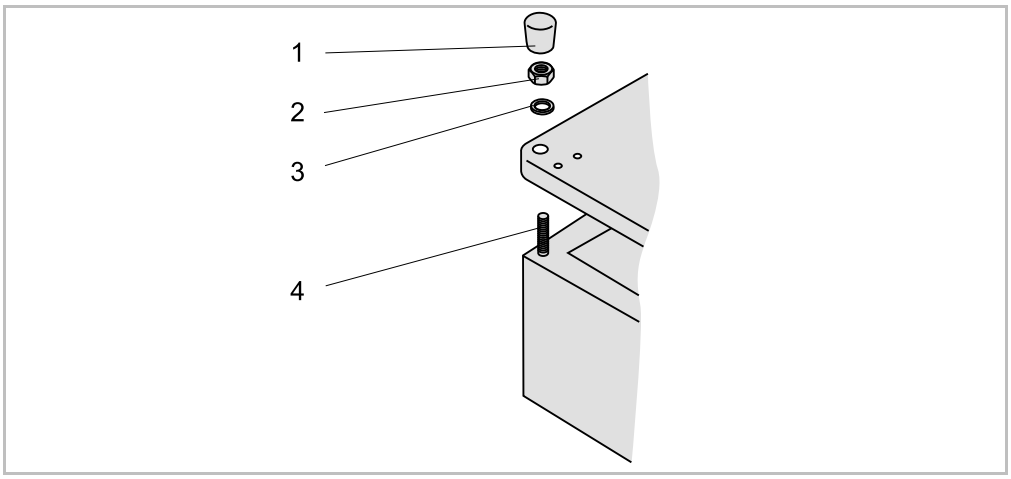


Fig. 4-8: Mounting the fan base plate onto the plinth
Pos. 1: Plastic caps
Pos. 2:Nuts
Pos. 3:Plastic washers
Pos. 4:Bolts

The plastic washers are used to seal gaps and to avoid damaging the paint on the base plate when tightening the nuts and to prevent corrosion.

Attention!
Cross tighten the threaded fasteners between the base plate and the plinth uniformly to avoid twisting.

4) Route the AC motor power supply and thermal protection cable or the EC motor control electronics cable through the base plate opening. The cables must be pulled through the PVC hose provided between the electric motor and the base plate, see Fig. 4-9 and Fig. 4-10.

The PG (PG 21) threaded bushing in the fan base plate provides a waterproof connection, meeting IP65. The top end of the hose is connected to the casing via a hole with the bushing.

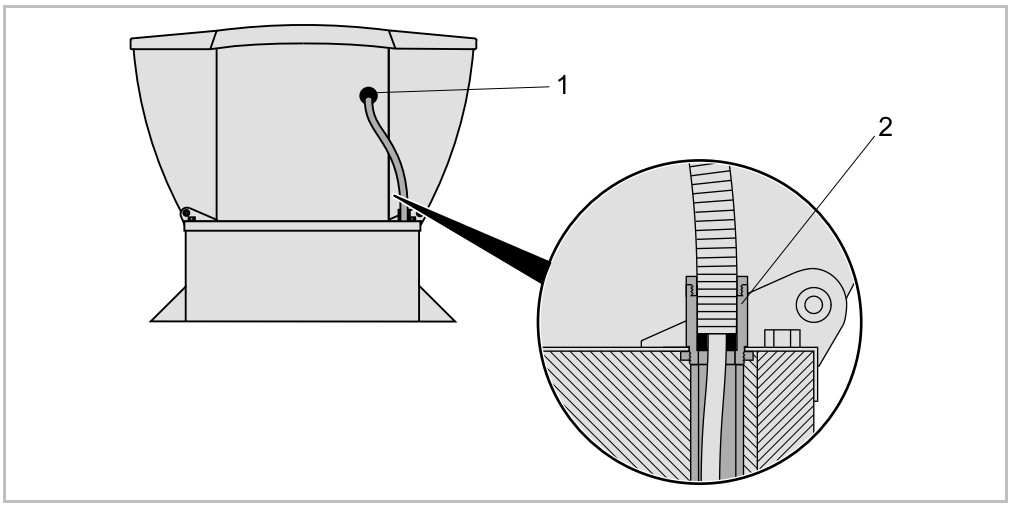


Fig. 4-9: Waterproof cable seal in the PVC hose
Pos. 1: PVC hose top end
Pos. 2: PG bushing

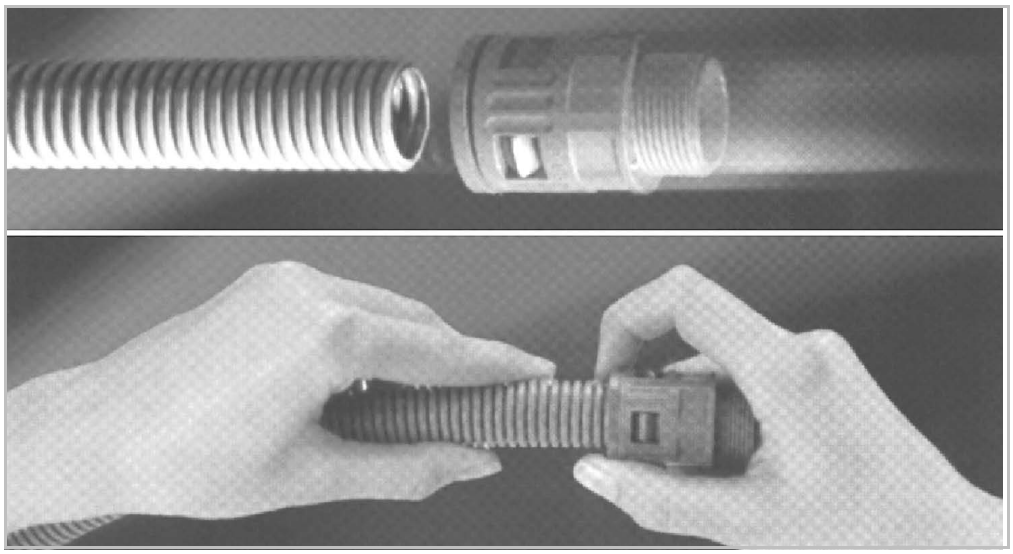


Fig. 4-10: Installing and removing the PVC hose with the PG bushing

5) Route the cables inside the fan casing and connect them to the fan switch or terminal block.

Attention!
The AC motor power supply and thermal protection cable or the EC motor control electronics cable must be sufficiently long.

4.2.5 Fan earthing

The fan base plate has an earth terminal.

- The fan must be bonded to a nearby metal structure and the building's lightning conductor using a conductor of adequate cross-section.

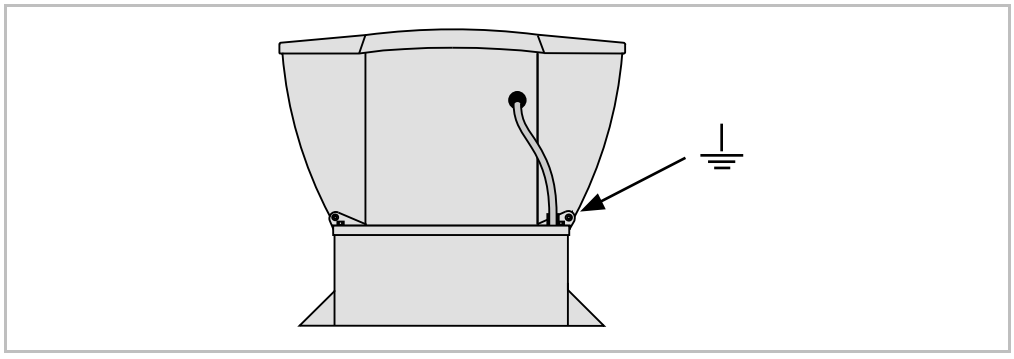


Fig. 4-11: Fan earthing

5 Electrical connections



Risk of electrocution!
The electrical installation of the fan may only be carried out by qualified licensed electricians pursuant to Clause 6 or higher of CUBP and CBU Regulation No. 50/78 Coll., following this operation manual and the applicable regulations:

- CSN standards
- Health and safety rules and regulations
- Installation instructions

5.1 Terminal block / fan switch wiring diagrams

The electrical connection of the roof fan and ductwork control unit is to be carried out according to the applicable wiring diagrams. Specific wiring diagrams are provided with the fan terminal block / fan switch.

5.2 Controller and fan connection

Roof fans are supplied with the electric motor connected to the terminal block or fan switch. The terminal block or fan switch is located under the fan cover.

- Mount the controller in its designated position (the controller is an accessory).
- Determine whether a single fan or a group of fans are to be connected.



Attention!
When connecting several fans, use a junction terminal block. The number of fans (up to 4 fans) per controller is determined by the controller's nominal rated current.



Attention!
To ensure the correct operating conditions, the EC fan motor control electronics are always to be supplied with mains power (i.e. energised) so that the control electronics enclosure is heated internally (to prevent condensation inside the enclosure). Fans with EC motors must therefore be switched on / off and operated by changing the 0-10V control voltage signal.

5.3 Cable connections

The cables (conductors) must be routed according to CSN standards.

The conductor type and cross-section is determined by an electrical engineering designer.

The motor cables and thermal protection cables (thermal contact or thermistor) must be routed separately to avoid interference.

Electrically shielded cables must be used for thermal protection conductors (thermal contact/thermistor), and electrical shielding must be connected in the control unit to the protective conductor terminal The decision whether to connect earthing at both ends can only be made on site (e.g. in case of severe interference), observing the applicable local regulations and codes.

If control is provided via a different system on site, fans can be provided with a thermal contact. The thermal contact can be connected directly to a 230 V source (up to 4 A) (e.g. to the contactor control coil). Overcurrent protection must be provided for electric motors.

After routing the cables, seal in place to make them waterproof.



Attention!
Tighten electric motor and terminal block bushings so that the cables cannot move.

5.4 Electric motor protection using thermal contacts / thermistors

If the fan is connected via a FläktGroup control unit, the thermal protection of the electric motor is connected (thermal contacts or thermistors). This connection provides the electric motor with overcurrent (thermal) protection. When coupled with FläktGroup controllers, the electric motor is switched off if the maximum permitted coil temperature is exceeded.

The warranty is limited if non-integrated thermal protection is used.

5.5 Power supply voltage fluctuation

According to CSN EN 60034-1 ed.2 a voltage tolerance of + / - 5% (zone A) applies to electric motors.

CSN IEC 60038 stipulates mains power supply voltage to be 230 V, 400 V and 690 V with a tolerance of up to + / -10%.

5.6 Fan speed control

5.6.1 Fan speed control by power supply voltage adjustment

The speed of all fan motors can be controlled by electronic voltage adjustment or transformers. However, if 3-phase flange mounted squirrel cage rotor motors are used, we recommend using a frequency inverter because of the torque.

In 3-phase, 2-speed fans it is possible to reduce speed to approx. 70% by switching from a delta connection (Δ) to a Y-connection (Y).

When using phase-fired speed control and depending on the installation, increased noise may be generated due to resonance. At speeds lower than 500 RPM we recommend using a sound attenuator.

5.6.2 Speed control using a frequency inverter

In drives with a frequency inverter, sinusoidal filters and input chokes must be used between the frequency inverter and the AC motor:

- When using several AC motors with one inverter
- For all fans with AC motors with an external rotor
- With mains voltage > 500 V
- With voltage peaks ts > 0.1µs at U ≤ 500 V



Risk of electrocution!
Do not confuse sinusoidal filters with motor filters (often also referred to as damping or du/dt filters).

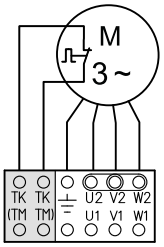
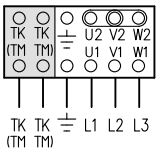


Attention!
The use of input chokes and sinusoidal filters limits current surges, additional noise and the risk of damage to the electric motor insulation.
Operation in a mains frequency fluctuation range (above 50 Hz) is not recommended.

5.7 Electrical connections

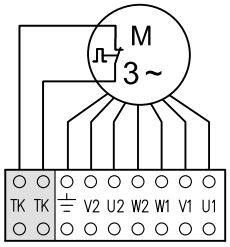
5.7.1 Flange mounted AC motors

Terminal block wiring diagram for 1-speed cage-rotor AC motors (3~ 400 V) (also for roof fans II 2G Ex h IIB+H_2 T4 Gb)

RJ##.####.##1# 	1-speed operation at operating voltage 3~ 400V using a 1-speed controller (MC301 / MC331) Power supply line: 3 + PE = 4 (5)-core cable Shielded line TC (TM): 2-core cable With thermal contacts / thermistors For operating voltage see the type plate
	**Conductor colour coding:** U_1,U_2 = orange V_1,V_2 = black W_1,W_2 = red

5.7.2 AC motors with an external rotor

Wiring diagram for 2-speed AC motors with an external rotor (3~ 400 V)

RJ##.####.##2# 	With thermal contacts, slip regulation Winding diagrams: Δ / Y For operating voltage see the type plate

The 2-speed three-phase external rotor AC motor can be used for 2-speed and 1-speed operation.

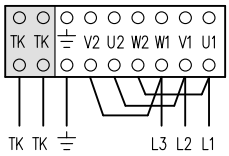
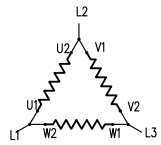
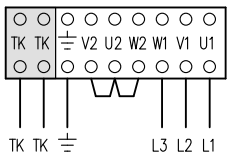
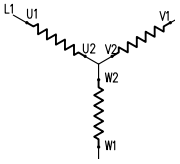
In order to use both speed ranges (high (Δ) and low (Y) speeds) during operation, a FläktGroup 2-speed switch must be used (the following diagram shows the AC motor terminal block connection for high (Δ) and low (Y) speed.

If only one speed is required (1-speed operation), a FläktGroup 1-speed controlled can be used.

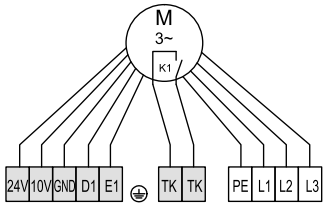
1-speed operation	**2-speed operation**
with a 1-speed FläktGroup controller (MC301 / MC331)	with a 2-speed FläktGroup controller (MC302 / MC332)
Power supply line: 3 + PE = 4-core cable	Power supply line: 6 + PE = 7-core cable
Shielded line TC: 2-core cable	Shielded line TC: 2-core cable

Special controllers are available on request for other operating voltages.

AC motor terminal block wiring diagram for 1-speed operation

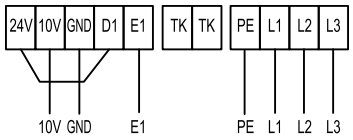
Motor connection	El. motor winding	Speed	Conductor colour coding
		High speed	U1 = brown U2 = red V1 = blue V2 = grey W1 = black W2 = orange
		Low speed	

Wiring diagram for EC motors with an external rotor, (3~ 400 V)

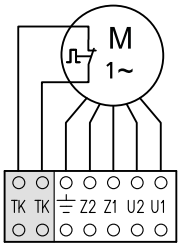
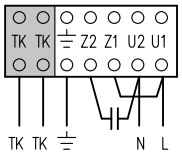
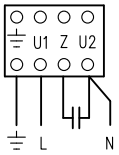
RJ##.####.##Z# 	With error contacts, continuous speed control For operating voltage see the type plate

Continuous speed control
Operation with a FläktGroup controller (OSH301 / OSH301Z) or potentiometer 950 EC1
Power supply line: 3 + PE = 4-core cable
Error detection line (error contacts): 2-core cable

EC motor terminal block wiring diagram for continuous speed control operation

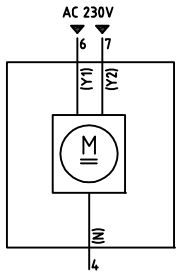
Motor connection	Speed	Control voltage
	continuous	10 V GND E1

Terminal block wiring diagram for 1-speed AC motors with an external rotor, (1~ 230 V)

RJ##.####.##4# 	1-speed operation at operating voltage 1~ 230 V with a 1-speed FläktGroup controller (MC301 / MC331) Power supply line: 2 + PE = 3-core cable Shielded line TC: 2-core cable With thermal contacts For operating voltage see the type plate
With wired TCs 	**Conductor colour coding:** U_1 = brown U_2 = blue Z_1 = black Z_2 = orange 1~ AC motor with capacitor and wired thermal contacts
Without TCs 	**Conductor colour coding:** U_1 = blue U_2 = black Z = brown 1~ AC motor with capacitor without wired thermal contacts

Attention!
The AC motor capacitor is already connected to the AC motor's terminal block and secured using a cable tie.

Wiring diagram for a damper with an actuator (RJZ#.36)

Wiring diagram	Conductor colour coding
	4 = blue 6 = black 7 = white

5.8 Electrical connections for a group of fans

For the parallel connection of several fans with a FläktGroup controller, use an intermediate terminal block. The power supply line and thermal contacts of electric motors can only be connected to a FläktGroup controller.

Example of the connection of a group of RJ#L and RJ#M fans with AC motors

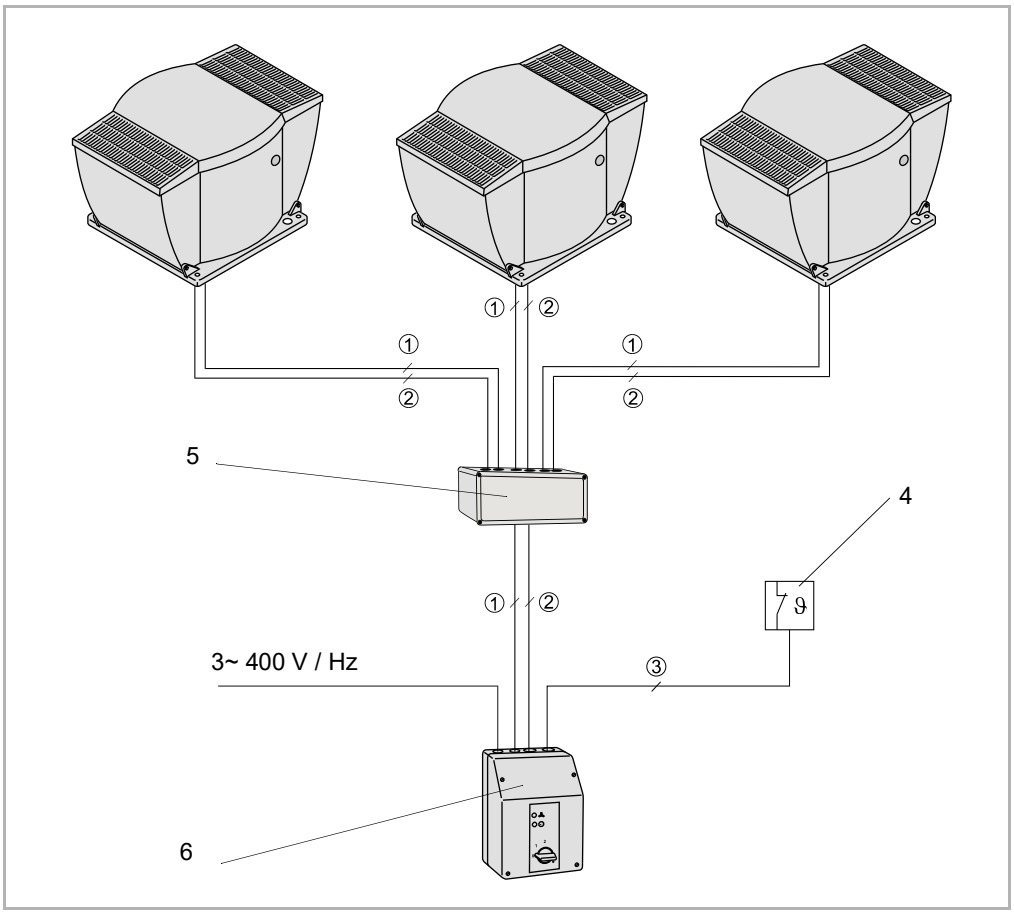


Fig. 5-1 A group of fans consisting of three RoofJETT roof fans with a FläktGroup controller and an intermediate terminal block

Pos. 1: AC motor power supply line: For the number of cores see the electric motor wiring diagram

Pos. 2: Shielded line: 2-core cable

Pos. 3: Sensor connection line (room thermostat): 2-core cable

Pos. 4: Sensor (room thermostat) (e.g. temperature, CO_2 etc. must be located in the vicinity of the fan)

Pos. 5: Intermediate terminal block

Pos. 6: FläktGroup controller MC301/MC331

Example of the connection of a group of RJ#Y fans with AC motors

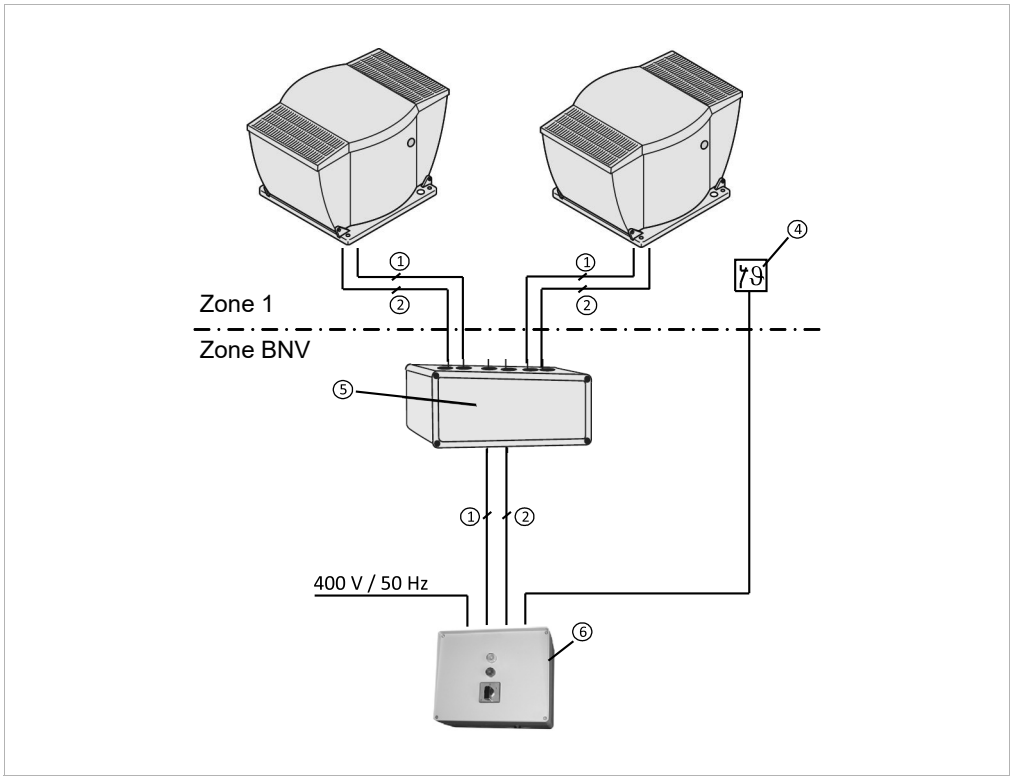


Fig. 5-2 A group of fans consisting of two RoofJETT roof fans with a FläktGroup controller and an intermediate terminal block

Pos. 1: AC motor power supply line: For the number of cores see the AC motor wiring diagram
Pos. 2: Thermal protection shielded line: 2-core cable
Pos. 3: Room thermostat connection line: 2-core cable
Pos. 4: Room thermostat for explosive environments - Type: 902014
Pos. 5: Intermediate terminal block - Type: 981860 (for installation in non-explosive environments)
Pos. 6: FläktGroup controller - Type: 986920 (for installation in non-explosive environments)

Example of the connection of a group of RJ#L fans with EC motors

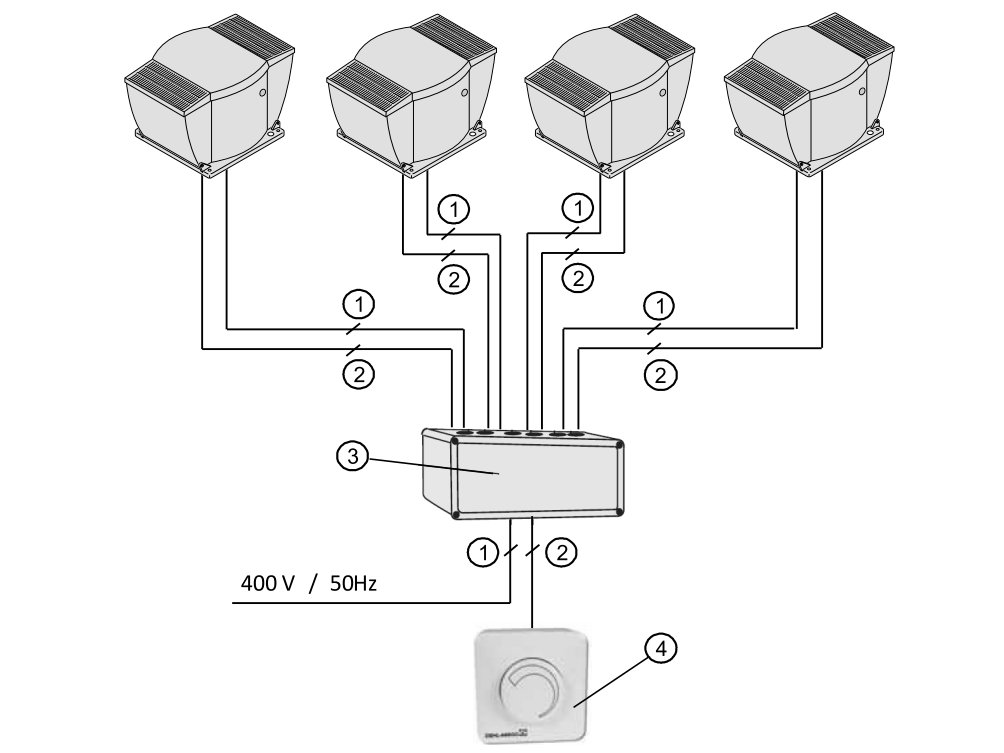


Fig. 5-3 A group of fans consisting of four RoofJETT roof fans with a FläktGroup controller with a potentiometer and an intermediate terminal block

Pos. 1: EC motor power supply line: 3 + PE = 4-core cable
Pos. 2: EC motor control connection (0 #. 10 V): 3-core cable
Pos. 3: Intermediate terminal block - Type: 981880
Pos. 4: Potentiometer 0-10 kΩ -Type: 950EC1

6 Commissioning

Attention!
Commissioning may only be carried out by trained personnel following this operation manual and the applicable regulations.

6.1 Safety inspection

Injury to persons!
If more than one roof fan is installed on the roof, their noise levels add up. When carrying out work in the vicinity of the fan, use ear defenders if noise levels exceed 85 dB (A).

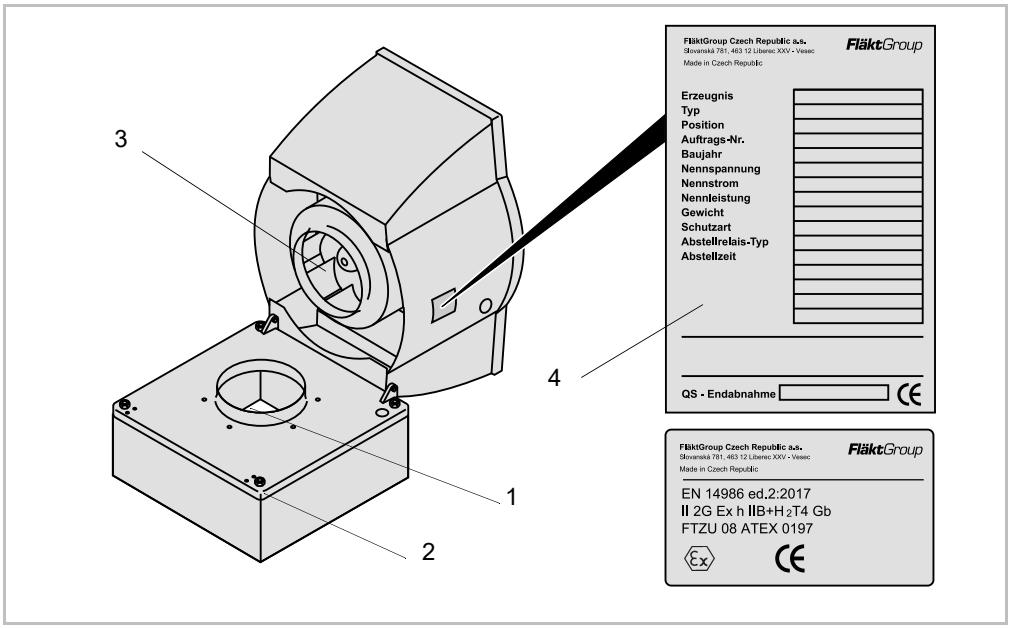


Fig. 6-1: Open fan casing during pre-commissioning inspection
Pos. 1: Air duct
Pos. 2: Nuts
Pos. 3: Fan impeller
Pos. 4: Type plate

Before commissioning, the following inspection must be carried out:

- Inspect the air duct and fan, removing any foreign objects (tools, small parts etc.). Free passage of air must be ensured (see Fig. 6-1, pos. 1).
- All safety features in place must be inspected (mechanical and electrical), (e.g. earthing resistance).
- Manually check the impeller for its freedom of rotation (see Fig. 6-1, pos. 3).
- Check the type of current, voltage and frequency of the mains power supply, making sure that it corresponds
  to the specifications of the fan or those on the type plate (see Fig. 6-1, pos. 4)
- Check the electrical connections inside the casing.
- Check the control features for proper function.
- Check whether all bolts, nuts etc. are tight and cannot come loose due to vibration (see Fig. 6-1, pos. 2).

6.2 Initial commissioning

6.2.1 Trial run

Following the above checks, a trial run should be carried out.

To turn the fan on, the fan switch should be in position “1”.

(fans with a fan switch are started by turning the switch to position “1”, on fans with a terminal block all terminals must be connected).

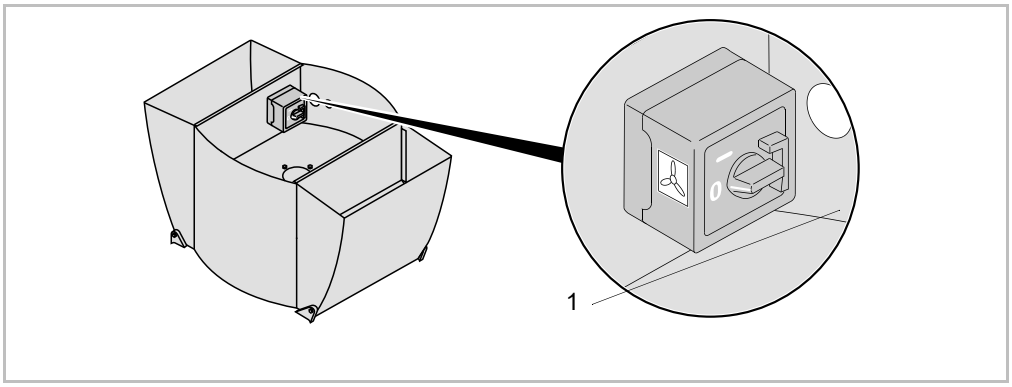


Fig. 6-2: Starting the fan
Pos. 1: Fan switch in position “0”

Risk of injury from rotating parts!
If a trial run is conducted when the fan casing is open, the impeller is not protected!
Never touch the impeller when the fan is running!

During a trial run check the impeller for the correct direction of rotation. The direction of rotation must correspond to the arrow indicator on the fan base plate.

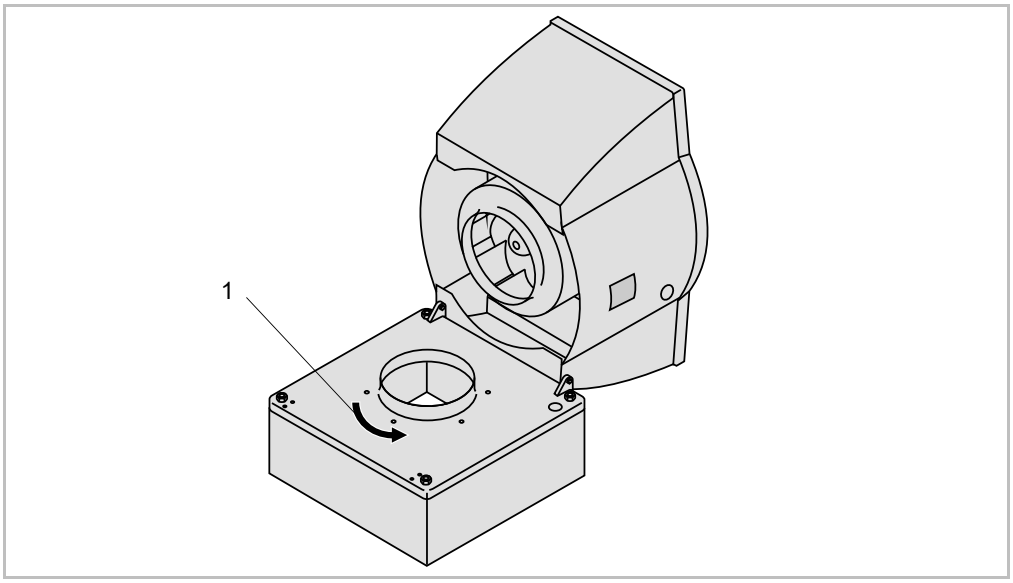


Fig. 6-3: Fan impeller rotation direction
Pos. 1: Fan impeller rotation direction arrow indicator

Attention!
If the direction of rotation is not correct, the fan’s capacity is significantly reduced.
It is necessary to change the poles of the AC motor, observing safety rules.
This is done by changing one phase on the fan switch or terminal block.

After a successful trial run, stop the fan using the fan switch or by disconnecting the terminals on the terminal block.

6.2.2 Initial commissioning procedure

1) After a successful trial run, turn on the fan using the remote control switch on the FläktGroup control unit (position “1”).

Attention!
On fans with a fan switch, the switch must be in position “1”, on fans with a terminal block all terminals must be connected.

2) After the operating speed is reached, measure the power consumption and compare the result with the electric motor power consumption value indicated on the motor or on the fan’s type plate. If the current is continuously higher, switch the fan off immediately. Check the ventilation function at all available speeds.

Attention!
When the current exceeds the value indicated for a prolonged period of time, the electric motor is automatically turned off by its thermal protection using thermal contacts/thermistors.

3) Check for the proper function of shut-off dampers.

Attention!
The shut-off dampers must open with the flow of air and remain closed when the fan is not running.

4) Check that the fan is not noisy when running. There must be no abnormal oscillation or vibration. Check that the electric motor does not generate any unusual noise.
5) Properly close (lock) the controller, fan and intermediate terminal block enclosure.
6) Hand over the unit in correctly.

Attention!
The maximum air flow rate V_{max} is indicated on the fan’s type plate.
The minimum air flow rate V_{min} should not drop below 30% of V_{max}.
On EC motor fans the control voltage should not drop below 4.5 V.

7 Operation

Attention!
When operating the roof fan, use the control unit to select the speed needed to provide proper ventilation (air extraction).
Roof*JETT* roof fans may only be operated in compliance with Commission (EU) Regulation No. 1253/2014.

The roof fans are normally operated (switched on and off) using a controller. In addition, a fan switch or terminal block must be used for activating and deactivating the fan during maintenance and inspection.

7.1 Switching the fan off

The roof fan is switched off using a controller.

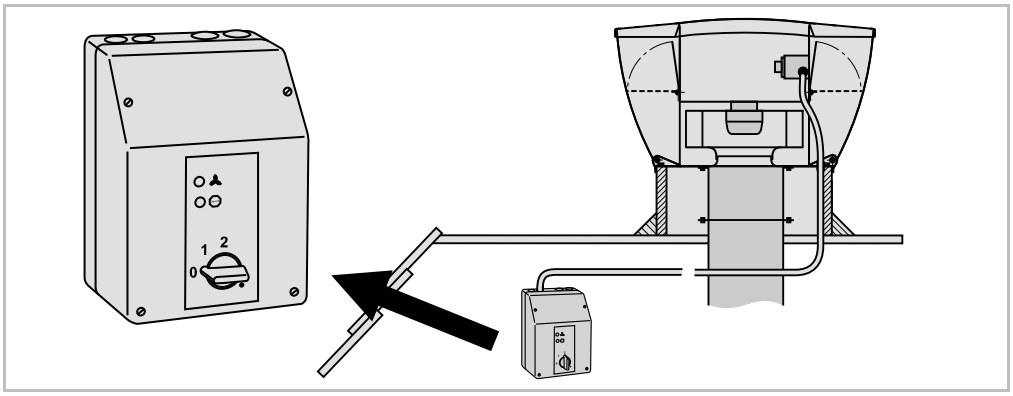


Fig. 7-1: Switching the controller to position “0”

The roof fan has either a fan switch or a terminal block.

Risk of electrocution!
If a fan has a terminal block, before conducting any work disconnect the electric fan motor from the power supply and secure it so that it cannot be unintentionally switched back on.

To prevent unintentional reconnection via the controller, it is necessary to:
- Set the fan switch to position “0”

or
- Disconnect and isolate the electric motor conductors in the terminal block. The terminal block casing must be closed.

In each case, the electric fan motor must be disconnected from the power supply and secured against re-connection before conducting any work.

7.1.1 Starting the fan after maintenance or repair

Before restarting the fan, run the safety checks indicated in Chapter 6.1.

Injury to persons!
Wear safety goggles for protection against flying parts!

- After completing work, fans with a fan switch can be restarted using this switch (position “1”); on fans with the terminal block, this is done by connecting the terminals. The fan is then started using a controller.

8 Maintenance and troubleshooting

Attention!
Maintenance may only be carried out by trained personnel following this operation manual and the applicable regulations.

Before conducting any work on the roof fan:



Risk of electrocution!
On fans with a fan switch turn the switch off and lock it to prevent an unauthorised restart!
On fans with a terminal block, switch the fan off using the control unit (controller) and secure it so that it cannot be unintentionally switched back on. Disconnect the conductors in the terminal block and insulate them.



Risk of injury from rotating parts!
Do not start work until the fan impeller has stopped completely and the electric motor has cooled down.

If maintenance is carried out regularly and when operating faults occur.

8.1 Regular inspection of RJ#L and RJ#M fans

As a minimum Roof*JETT* RJ#L and RJ#M fans, including accessories, must be inspected once every 3 years. The period depends on operating conditions.
The following steps should be taken as part of the regular inspection:

- Check the fan impeller for regular operation, check the heating and noise of the electric motor.
- Remove any dust and dirt build-up.
- Check there is no damage to the roof fan.
- Remedy any defects found.
- Complete the inspection report in this operation manual (see page 45).

8.2 Regular inspection of RJVY fans

As a minimum Roof*JETT* RJVY fans, including accessories, must be inspected once every 3 years. The period depends on operating conditions.
The following steps should be taken as part of the regular inspection:

- Check the fan impeller for regular operation, check the heating and noise of the electric motor.
- Remove any dust and dirt build-up.
- Check there is no damage to the roof fan.
- Check the fan impeller mounting and the blade clearance between the fan impeller (Pos. 2,Abb. 2-6, page 16) and the fan base plate (Pos.1,Abb. 2-6, page 16). The minimum clearance is indicated in Table 2-3 on page 15.
- Remedy any defects found.
- Complete the inspection report in this operation manual (see page 45).

8.3 Operational faults

Deviation from normal fan operation will lead to malfunctions and must be checked by maintenance personnel.



Damage to the fan!
Long-term faults may lead to damage to the fan and personal injury.

The following table is intended to provide maintenance personnel with guidelines regarding possible faults and their rectification.

Fault	Cause	Troubleshooting
The fan is not running, the control box (controller) is on	The fan switch is in position “0” or conductors are not connected in the terminal block.	Set the fan switch to position “1” and connect the conductors in the terminal block.
The fan is not running, the controller is on, the fan switch is on / terminals connected	a) No power supply or control protection circuitry fault	Check the power supply voltage and fuses, check connections
	b) Electric motor protection circuitry tripped	Check the temperature of the electric motor, alternatively let it cool down and start it again[1)] (if the fault occurs repeatedly, remove the cause of the electric motor overheating).
Electric motor protection circuitry trips during ventilation or trial runs		Check the fan impeller for its freedom of rotation, damage during storage, electric motor winding faults, power supply lines.
The fan impeller slips		Check the fan impeller area for foreign objects, check its attachment to the electric motor.
The fan is running but no or little extraction occurs	a) Blocked air flow	Provide for free air flow, clean the fan impeller, check ductwork, check the function of shut-off dampers.
	b) Wrong direction of rotation	Change the direction of rotation by changing the wiring poles (this only applies to AC motors).
Water leaks from the outside to the space under the fan	a) Plinth insulation not carried out properly	Repair the insulation according to this operation manual.
	b) Shut-off dampers remain open even when the fan stops.	Release and clean the dampers so that they function properly.

Table 8-1: Faults and troubleshooting

1) Starting after removing a fault: First, select position “0” on the fan speed controller, then set it again to the required speed level.

Attention!
Failure to carry out regular checks will invalidate the manufacturer’s warranty in the event of a fault.

If a fault cannot be remedied by maintenance personnel, please contact the FläktGroup servicing department.

9 Dismantling and disposal



Damage to the environment!
Roof*JETT* roof fans may only be dismantled and disposed of by an authorised specialist personnel!

9.1 Dismantling

Dismantle a Roof*JETT* roof fan as follows:



Risk of electrocution!
Before starting any work, disconnect the fan from the power supply to prevent electric shocks. Ensure that the unit is isolated from the power supply and secured against being switched back on at an appropriate point of the on-site power supply.



Injury to persons!
Secure the roof fan against sliding. Comply with the transport guidelines.

9.2 Recycling



Recycling
The disposal of the Roof*JETT* fan and its components must be carried out by an authorised specialist firm. This specialist firm must make arrangements for the following:

- parts must be sorted according to the type of material,
- operating media must be separated and sorted according to their properties.



Damage to the environment!
All parts and operating media must be disposed of in a suitable, environmentally friendly manner, in compliance with local laws and regulations.

The disposal of the unit and its components must be carried out by a authorised specialist firm. This firm must make arrangements for the following:

– Separation of the parts of the unit according to material type
– Separation of the operating media according to their properties
– Compliance with local and international rules and regulations regarding material disposal:
  – Waste Electrical and Electronic Equipment Directive (WEEE Directive) 2012/19/EU
  – Directive on Waste 2008/98/EC

Roof*JETT* roof fan .....

(the type to be added at 1st inspection)

RJZ accessories .....

(the type to be added at 1st inspection)

.....

.....

1st inspection carried out by ..... (date, stamp, signature)

2nd inspection carried out by ..... (date, stamp, signature)

3rd inspection carried out by .....(date, stamp, signature)

4th inspection carried out by .....(date, stamp, signature)

5th inspection carried out by .....(date, stamp, signature)

6th inspection carried out by .....(date, stamp, signature)

7th inspection carried out by .....(date, stamp, signature)

8th inspection carried out by .....(date, stamp, signature)

9th inspection carried out by .....(date, stamp, signature)

10th inspection carried out by .....(date, stamp, signature)

FläktGroup

EC DECLARATION OF CONFORMITY

Pursuant to European Parliament and Council Directive 2014/34/EU, European Parliament and Council Directive 2014/30/EU and European Parliament and Council Directive 2006/42/EC this is an original EU declaration of conformity, No. 2020/071/5AB12012

Manufacturer:
FläktGroup Czech Republic a.s., Slovanská 781, 463 12 Liberec XXV – Vesec, Czech Republic, IC (Company ID): 46708375

Entity authorized to compile technical documentation:
FläktGroup Czech Republic a.s., Slovanská 781, 463 12 Liberec XXV – Vesec, Czech Republic, IC (Company ID): 46708375

Product description

Roof fans

RoofJETT RJVY ....

RJVY type RoofJETT explosion-proof roof fans, with separately approved explosion-proof electric motors, are designed for the extraction of flammable gases and vapours from environments with a risk of explosion, ZONE 1. RJVY series fans use explosion-proof electric motors in "a flameproof enclosure" configuration II 2G Ex h IIB + H2 T4 Gb. RJVY series fans are available in 4 sizes for air flow rates from 1,485 to 9,100 m^3/h for max. static pressures of up to 470 Pa. Fans can include RJZX series accessories.

Identification:

CE Ex II 2G Ex h IIB + H_2 T4 Gb

ALL APPLICABLE PROVISIONS THE PRODUCT IS IN COMPLIANCE WITH:
- European Parliament and Council Directive 2014/34/EU (government Regulation 116/2016 Coll.)
- European Parliament and Council Directive 2014/30/EU (government Regulation 117/2016 Coll.)
- European Parliament and Council Directive 2006/42/ES (government Regulation 176/2008 Coll.)

REFERENCE TO TECHNICAL STANDARDS AND SPECIFICATIONS:
CSN EN ISO 80079-36:2016; CSN EN ISO 80079-37:2015; CSN EN 14986 ed.2:2017; CSN EN ISO 14120:2017; CSN EN ISO 3746:2011; CSN EN ISO 12100:2011; CSN EN ISO 11202:2010; CSN EN 60204-1 ed.2:2007; CSN EN ISO 13857:2008; CSN EN 62233:2008; CSN EN 55014-2 ed.2:2017; CSN EN 55014-1 ed.3:2007; CSN EN 61000-3-2 ed.4:2015; CSN EN 61000-3-3 ed.3:2014; CSN EN 61000-6-3 ed.2:2007

METHOD OF CONFORMITY ASSESSMENT
Conformity assessment pursuant to European Parliament and Council Directive 2014/34/EU (government Regulation 116/2016 Coll.):

Type test certificate	number:	FTZU 08 ATEX 0197 + Appendix 1 + Appendix 2 + Appendix 3 + Appendix 4
	Validity:	from: 21/07/2008 (01/05/2012, 29/05/2013, 05/06/2013, 31/8/2016, 14/5/2018) to: 31/01/2025

Issued by:	Fyzikálně technický zkušební ústav (Physical-Technical Testing Institute), NB 1026, Pikartská 7, 716 07 Ostrava, CR; ICO (Company ID) 577 880

Manufacturer FläktGroup Czech Republic a.s. was awarded a certificate by TÜV Rheinland/Berlin-Brandenburg pursuant to CSN EN ISO 9001:2016

CONFIRMATION: Manufacturer FläktGroup Czech Republic a.s. confirms that the properties of the products comply with the basic requirements of European Parliament and Council Directives 2014/34/EU, 2006/42/EC and 2014/30/EU and the technical standards and regulations indicated above. They are safe under the conditions of standard use as defined by the manufacturer. The manufacturer adopted measures to ensure compliance of the products being introduced into the market with the technical documentation and basic requirements. The manufacturer took all necessary measures to ensure that the production process, including final inspection and end product tests, guarantees production uniformity and the compliance of the products with the types described in the certificate and with applicable basic requirements.

The manufacturer declares that measures were applied to the product to prevent explosive atmosphere initiation by eliminating all electrical and non-electrical sources of initiation in terms of Article 1.0.1. b) Appendix 2 NV No. 116/2016 Coll. (Directive 2014/34/EU), that the product complies with all requirements for classification in category II 2G, conforms to the technical documentation and the basic requirements of applicable government regulations and may be operated in explosive environments.

Issued in Liberec on: 01/ 02/ 2020

Name, position: Ing. Eduard Horbal', Chairman of the Board

Signature

EC DECLARATION OF CONFORMITY

pursuant to Directive 2006/42/EC of the European Parliament and of the Council (government regulation No. 176/2008 Coll.) (original EC Declaration of Conformity) 2020/030/5AB15513

Manufacturer:
FläktGroup Czech Republic a.s., Slovanská 781, 463 12 Liberec XXV - Vesec, Czech Republic, IC (Company ID): 46708375

Entity authorized to compile technical documentation:
FläktGroup Czech Republic a.s., Slovanská 781, 463 12 Liberec XXV - Vesec, Czech Republic, IC (Company ID): 46708375

Description and identification of machinery: Roof*JETT* roof fans
RJ##.####.#### including accessories with control systems and controllers

Roof*JETT* roof fans RJ## are designed for ventilation in residential and industrial buildings, the extraction of air from production premises and exhaust of smoke products in the event of fire. The fans have a radial design and consist of a base plate, a casing made of galvanized painted sheet steel, a fan impeller made of galvanized sheet steel or aluminium and a flange mounted electric motor. The base casing has automatic aluminium dampers which prevent spontaneous airing of premises to be ventilated and the penetration of air from the outside when the fan is not running. The fan is produced in three design lines: RJV#, RJN#, RJS#. The RJV# series is a radial roof fan with vertical exhaust airflow, with integrated automatic shut-off dampers and a tiltable fan casing. The RJN# is identical to the RJV# series but has no tiltable fan casing. The RJS# series is like the RJV# series but includes a sound damping fan casing. The fan is produced for power supply voltage 1x230 V AC, 50 Hz or 3x400V AC, 50 Hz.

Declaration:
The machinery complies with all relevant provisions of Directives 2006/42/EC, 2014/30/EU and 2014/35/EU.

List of harmonized standards applied in conformity assessment:
CSN EN ISO 14120:2017, CSN EN ISO 3746:2011, CSN EN ISO 12100:2011, CSN EN ISO 11202:2010,
CSN EN 60204-1 ed.3:2019, CSN EN ISO 13857:2020, CSN EN 62233:2008, CSN EN 55014-2 ed2:2017

This declaration applies exclusively to the machinery in the condition in which it is being introduced into the market and does not include parts added by the end user at a later date or subsequent modifications made by the end user.

Issued in Liberec on: 1/ 7/ 2020

Name, position: Ing. Eduard Horbal', Chairman of the Board



Signature

Year of manufacture:

FläktGroup

FG_DC-2010-0048-GB_RoofJETT_OM_R9-06-2022 © Copyright 2022 FläktGroup

EXCELLENCE IN SOLUTIONS

FläktGroup is the European market leader in energy-efficient solutions for air handling applications suitable for every application area according to your requirements. We offer our customers innovative technologies, high quality and outstanding performance supported by more than a century of accumulated industry experience. Our wide product range and market presence in 65 countries worldwide, guarantee that we are always by your side, ready to deliver excellent solutions.

PRODUCT FUNCTIONS BY FLÄKTGROUP

Air Treatment | Air Movement | Air Diffusion | Air Distribution | Air Filtration
Air Management | Air Conditioning & Heating | Controls | Service

*» Learn more on our website **www.flaktgroup.com** or contact your local sales office.*